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SURVEY

Semantic Communication Empowered 6G Networks: Techniques, Applications, and Challenges

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ABSTRACT With the explosion of intelligent network applications and the prosperity of artificial intelligence (AI) technologies, the sixth generation (6G) wireless networks are not only expected to further improve network capacity, but also anticipated to establish a new architecture of "Intelligent Connectivity of Everything". Semantic communication (SC) is a promising solution for future 6G networks due to its natural capability of integrating application requirements and information meaning into data transmission processes. In this paper, a comprehensive survey that overviews how SC can be applied for 6G networks and the key technologies of SC is presented. For this purpose, we first provide a detailed overview of the concepts of semantic information (SI) and SC, as well as the classifications of SC. Then, we present the vision of mutual support between SC technologies and 6G networks, as well as the potential benefits using SC for 6G applications. To achieve the benefits of SC, the fundamental theories and four important technologies in SC are thoroughly investigated, which are SC system architecture design, SI extraction, SI transmission, and SC performance evaluation. Then, we introduce open problems and potential research directions pertaining to SC. In a nutshell, this paper provides a holistic review of concepts, applications, fundamentals, key technologies, existing challenges, and open issues of SC tailored to the requirements of 6G networks.

INDEX TERMS Semantic communication (SC), semantic information (SI), 6G networks, artificial intelligence (AI), knowledge graph (KG).

I. INTRODUCTION

The future 6G networks aim to provide data intensive and resource intensive network services, such as augmented reality and digital twins, that are centered around the perceptual and cognitive requirements of intelligent agents [1]. However, current communication systems is approaching the Shannon Limit by using advanced encoding and modulation techniques. Communication systems aimed at precisely transmitting and recovering bitstreams are becoming progressively difficult to improve quality and experience of network services. Based on the artificial intelligence (AI) technologies, semantic communication (SC) systems enable intelligent agents to understand and transmit the meaning of information to satisfy application requirements. As an emerging communication paradigm, SCs that can improve communication efficiency and support intelligent network services have attracted much attention.

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A. MOTIVATION

1) REQUIREMENTS FOR FUTURE 6G NETWORKS

Cellular mobile communication systems have evolved from the first generation (1G) to the fifth generation (5G). Almost each generational evolution involved using new frequency bands and implementing new physical layer technologies to increase network capacity [2]. In June 2023, ITU-R WP5D completed the recommendation framework for IMT-2030 (Global 6G Vision) [3]. This framework expands three

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scenarios of 5G networks and defines six major scenarios for 6G networks, which are immersive communication, hyper reliable and low-latency communication, massive communication, ubiquitous connectivity, integrated AI and communication, and integrated sensing and communication. For scenarios of immersive communication as well as integrated sensing and communication, 6G networks need to provide a data rate of at least 2.0 Tbps [4]. However, wireless network spectrum resources are limited and the capacity of the current communication system is approaching the Shannon Limit. Meanwhile, the large amount of redundant information contained in the big data generated by massive sensors will lead to a significant waste of wireless resources. For scenarios of massive communication and ubiquitous connectivity, large-scale devices using different protocols and incompatible data formats request access to 6G networks. However, traditional communication technologies cannot directly connect such heterogeneous devices and support the interoperability between such devices. For scenarios of integrated AI and Communication as well as hyper reliable and low-latency communication, the requirements of interaction among intelligent agents explode. However, artificial machine interaction languages are rigid and inefficient, especially for dynamic and self-organizing tasks for intelligent agent swarms. To improve communication efficiency and support intelligent network applications, a revolutionary communication technology called semantic communication (SC) is proposed. SC technologies enable the transmitter to accurately transmit the meaning of the information, and enable the receiver to correctly understand the intentions of the transmitter. Based on SC technologies, the 6G network vision of transitioning from interconnection of everything to intelligent interconnection of everything can be realized.

2) AWARENESS OF COMMUNICATION NATURE

In 1948, Claude Shannon stated that the fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point, and then proposed Shannon information theory. In Shannon information theory, semantic aspects of communication were identified but considered irrelevant to the technical communication problems. The reason is that the meaning of a message may be related to "certain physical and conceptual entities" and involving meaning in mathematical models might affect the universality of the theory [5]. However, in addition to accurately conveying the message, conveying the meaning of the message and achieving the effect of the information are also within the scope of communications. In 1953, Warren Weaver proposed three levels of communication problems that are accurate symbol transmission, semantic information exchange, and effectiveness of message exchange [6]. Communication systems derived from Shannon theory focus on the first-level technical problem and aim to ensure the accuracy of received symbols. After decades of development in communication technologies, the technical problem of how to effectively turn unreliable channels into reliable channels has been basically solved. However, with the definition of 6G intelligent applications, the semantic problem, which used to be considered as a "secondary" problem, has become an urgent issue to be solved.

3) BREAKTHROUGHS IN AI AND COMPUTING POWER

With the breakthroughs in AI technologies and the improved computing power due to computer hardware upgrades, communication devices are expected to intelligently process messages and AI based SC systems can be implemented. The future 6G networks are expected to achieve ubiquitous wireless intelligence and, hence require to support massive access devices and high traffic data. In order to reduce the pressure of data transmission on spectrum resources, communication devices can utilize AI algorithms to eliminate redundant information from multiple information sources, while transmitting specific information to meet application requirements. Therefore, SC is a key technology to realize the trade-off between computational and communication overheads in the context of spectrum congestion as well as the explosion of AI technologies.

B. PREVIOUS WORKS

The exciting new opportunities for using SC in wireless networks purposes have spawned a number of interesting surveys such as in [7], [8], [9], [10], [11], [12], [13], and [14]. The work in [7] provided an overview of the works on deep learning (DL) based and end-to-end (E2E) SC systems. In [8], the authors first analyzed the fundamental theory and mathematical framework of SC, and then summarized four implementation approaches of SC, including semantic information theory based SC, knowledge graph (KG) based SC, DL based SC, and goal-oriented SC. However, the works in [7] and [8] did not introduce the metrics for evaluating the performance of SC systems. The authors in [9] comprehensively investigated the performance evaluation metrics for wireless SC systems, optical SC systems, and quantum SC systems. The works in [10] and [11] reviewed the existing works and challenges of SC systems for multimodal data (i.e., texts, images, and videos) transmission, as well as introduced the robustness, accuracy, and security assurance technologies for multimodal data SC systems. However, the works in [7], [8], [9], [10], and [11] ignored to analyze the potential applications of SC in future 6G networks. In [12] and [13], the authors respectively reviewed the mathematical foundations and three key technologies, e.g., semantic information (SI) extraction, SI transmission, and evaluation metrics of SC systems for 6G networks. The work in [14] introduced SC empowered applications, outlined three key technologies of SC including SI extraction, and semantic segmentation, and semantic coding, and analyzed the problems of existing SC works in terms of efficiency, robustness, adaptability, and reliability. However, existing works in [12], [13], and [14] reviewed parts of key technologies in complete SC processes,

TABLE 1. A comparison between existing surveys and our survey.

	Technologies					Performance Evaluation	Applications
	SI Extraction	SI Encoding	SI Fusion	SI Transmission	Architecture of SC		
[7]					✓		
[8]	✓	✓		✓	✓		
[9]						Metrics for wireless, optical, and quantum SC systems.	
[10]	✓	✓	✓	✓		Robustness, accuracy, and security of SC systems.	
[11]	✓	✓	✓	✓		Security of SC systems.	
[12]	✓			✓			Intelligent transportation, distributed networks, and UAV networks.
[13]	✓	✓		✓	✓		Brain-to-computer interaction, holographic communications, multi-sense XR, and affective computing.
[14]	✓	✓				Efficiency, robustness, and reliability of SC systems.	UAV networks, remote sensing, intelligent transportation, and intelligent healthcare.
Our survey	✓	✓	✓	✓	✓	Semantic channel capacity, task oriented metrics, semantic perception metrics, and SC system metrics (e.g., timeliness, energy efficiency, security, robustness, and adaptability).	6G major scenarios, i.e., massive communication, ubiquitous connectivity, integrated AI and communication, integrated sensing and communication, immersive communication, and hyper reliable and low-latency communication.

as listed in Table 1. Moreover, the surveys such as [12], [13], and [14] mainly limit their discussions to existing network applications, rather than provided a comprehensive analysis of the feasible applications of SC in future 6G networks. A detailed comparison of our paper and the existing surveys is listed in Table 2.

C. CONTRIBUTIONS

The main contribution of this paper is to provide a comprehensive survey on SC technologies for major scenarios of future 6G networks, especially SI extraction, SI transmission, and SC performance evaluation technologies. The overarching goal is to overview the emerging research contributions of SC that address the major opportunities and challenges in developing SC enabled intelligent wireless systems in 6G networks. To the best of our knowledge, this is the first survey that gathers the state-of-the-art research contributions related to the connotations, applications, theoretical foundations, as well as SI extraction, SI transmission, and SC performance evaluation technologies from the vision of future 6G networks. Our main contributions include:

- We present a vision of mutual support between SC technologies and 6G networks. In particular, we analyze the potential advantages of using SC in 6G networks and provide example use-cases for each 6G scenario. Moreover, we analyze how services (e.g, edge intelligence) provided by 6G networks can facilitate the implementation of SC.
- We provide a comprehensive overview of three key technologies involved in the SC architecture, which are SI extraction, SI transmission, and SC performance evaluation technologies. In terms of the SI extraction, we introduce DL based SI extraction, KG based SI extraction, and HST based SI extraction approaches, and elaborate on how they apply to different data modalities and accomplish different SC tasks. In terms of the SI transmission, we review the existing SI transmission technologies, including SI filtering, adaptive SI transmission, as well as resource allocation and multiple access solution for SC networks. In terms of the performance evaluation of SC systems, we investigate the metrics of semantic channel capacity, task oriented metrics, semantic perception metrics, and SC system metrics including timeliness, energy efficiency, security, robustness, and adaptability.
- We analyze seven open issues, covering the fundamental theory, system architecture, and three key technologies of SCs.

The scope of the survey is organized as follows, as shown in Fig. 1. We introduce the concepts related to SC in Section II, which pave the way for the later exposition. In Section III, we provide the motivating applications of SCs in future 6G networks. Based on the requirements of such applications, we first analyze the fundamental theories and system architectures of SC in Sections IV-A and IV-B, respectively. The architectures of the existing SC system include three key technologies, which are SI extraction, SI transmission, and SC performance evaluation. We specify these three key technologies in Section IV-C, IV-D, and IV-E, respectively. Based on the challenges and state-of-the-art of

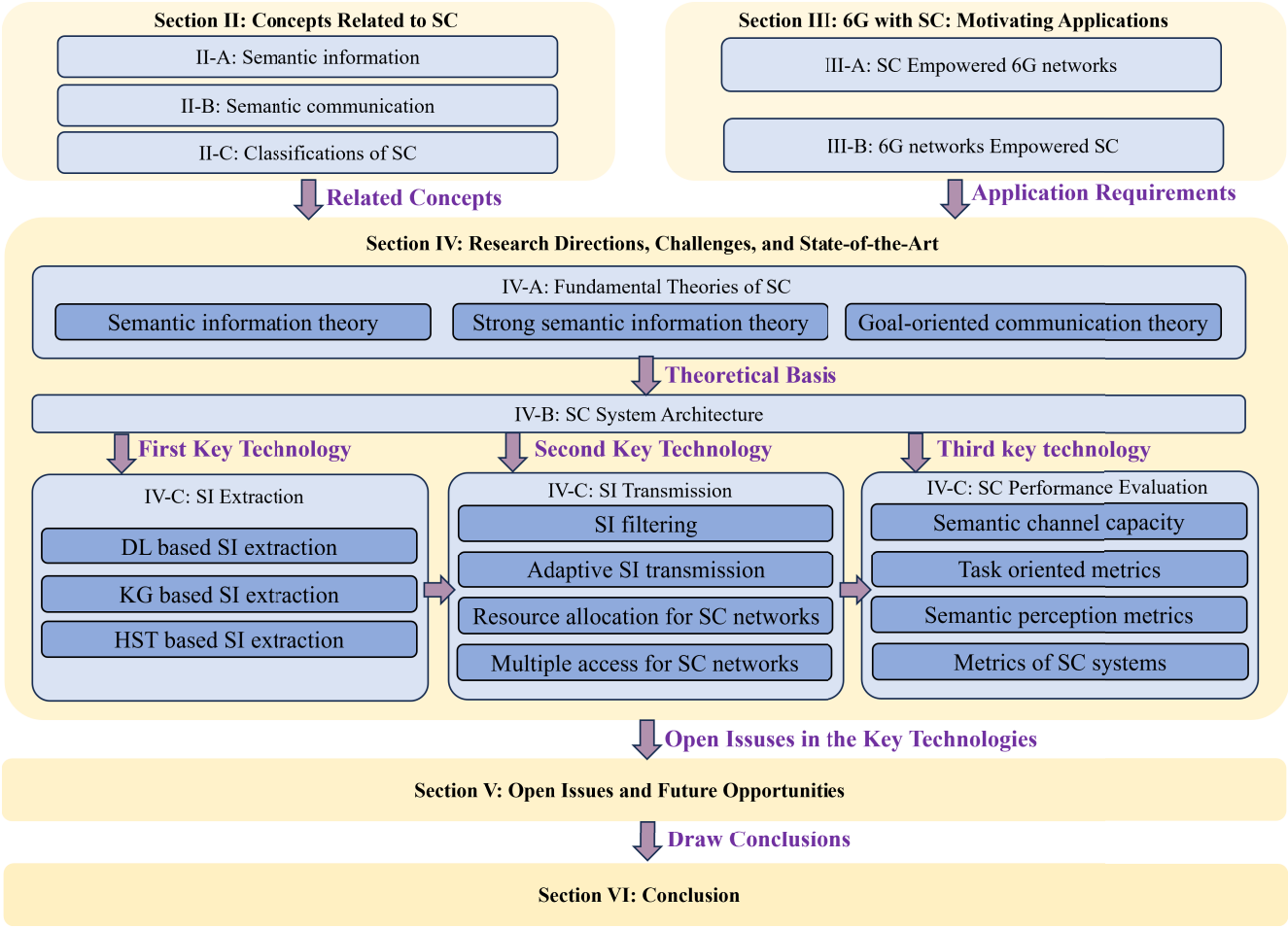

FIGURE 1. Structure of the survey.

each research direction, we identify a series of open issues for future research in Section V. Finally, conclusions are drawn in Section VI.

II. CONCEPTS RELATED TO SC

A. SEMANTIC INFORMATION

Information can be expressed in three levels that are syntactic information, semantic information, and pragmatic information [15]. Here, semantic information refers to the meaning of the information that can be used to eliminate uncertainty. Wikipedia gives the sources of SI as "any kind of meaningful information provided by language, text, symbols, data, formulas, theories, and so on". In [15], the authors explain SI from objective and perceptual perspectives. From an objective perspective, SI is a series of objective descriptions of things, such as their objective properties, associations, and hierarchical structures. From a perceptual perspective, SI is the perceived information of the subject towards the object, and can be described as the joint mapping of syntactic information and pragmatic information in the semantic spaces. Here, pragmatic information is the subjective purpose of receiving information. For example, the SI of the image

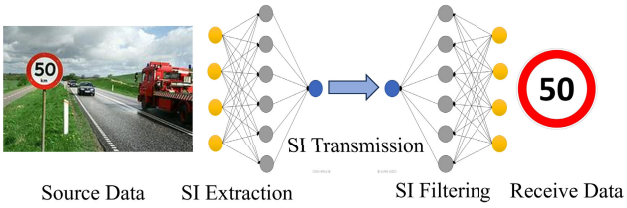

FIGURE 2. An example of SI.

in Fig.2 from an objective perspective can be given as "*[vehicle1, color, Black],[vehicle1, distance, 0.1KM],[street, number of lanes, 2][traffic sign, content, speed limit 50],...*". We assume that the pragmatic information is vehicle speed control in autonomous driving. The SI of Fig.2 from a perceptual perspective can be obtained as "*[vehicle1, distance, 0.1KM],[traffic sign, content, speed limit 50]*".

B. SEMANTIC COMMUNICATION

In recent years, SC as a new communication paradigm has rised widespread attention. In [16], the authors pointed out

that SC is a new task-oriented communication technology for intelligent agents, which provides the possibility to derive task-oriented machine languages and improves the collaboration efficiency between intelligent agents. The work in [16] also claimed that one of the key issues of SC is to find suitable semantic representation and appropriate coding schemes to to ensure that SI is transmitted in a way that minimizes ambiguity and redundancy. In [12] and [17], the authors believed that SC is an AI based communication technology that integrates user and application requirements and the meaning of the information into data processing and transmission. The authors in [7] stated that the core of SC is to extract the meaning of information at the sender and successfully interpret the meaning at the receiver with the help of the matching knowledge base (KB) between the sender and the receiver.

Existing works [12], [18], [19], [20] analyzed the relationship between SC systems and traditional communication systems. The work in [12] indicated that SC enables the transmitter to understand and transmit the information represented by semantic symbols, from which the receiver can access the meaning of the information. In contrast, traditional communication systems transmit symbol sequences and leave the task of understanding the information within the symbols to the receiver. Similarity, the authors in work [18] summarized that SC enables the receiver to draw conclusions from the received SI and its local KB, rather than mechanically decoding the message. The works in [19] argue that, different from traditional communications that focus on the accurate transmission of symbols, SCs pay more attention to the information requirements of the receiver and the interpretation of the received information. In [20], the authors pointed out that SC can be implemented based on the classic Shannon communication system by integrating AI models into physical layers of the communication agents. Based on the descriptions of SC in existing works, the characteristics of SC that differ from traditional communication systems are summarized as follows:

- Communication agents are intelligent and can support understanding information before transmission. SC technologies shift the communication paradigm from "transmission before understanding" to "understanding before transmission".
- The transmitter and receiver of a SC system maintain a matching prior KB that enables them to have a consistent understanding of SI and the goal of communications.
- Different from traditional communication systems that are designed to accurately transmit original data, the design of SC systems, such as waveform design, protocol design, and resource allocation, aims to achieve the goal of communications by accurately conveying the meaning of the information.

C. CLASSIFICATIONS OF SC

According to the three levels of communications identified by Weaver and Shannon [6] and the role of SC in applications, the existing works related to SC can be divided into three categories, which are semantic-oriented SC, goal-oriented SC, and semantic-aware SC, as done in [12]. The authors in [12] introduced the details of each type of SC. To achieve semantic-oriented SC, the works in [15] and [18] and the works in [21], [22], and [23] studied semantic information theory-based SC systems and AI-based SC systems, respectively. To achieve semantic-oriented SC, lots of existing works such as in [24], [25], and [26] proposed AI-based SC systems. To achieve the semantic-aware SC, the works in [27], [28], [29] proposed AI-based SC systems.

The differences between semantic-oriented SC, goal-oriented SC, and semantic-aware SC can be given as follows:

- *In terms of SI extraction,* the SI extracted by semantic-oriented SC systems contains complete objective information of source data, i.e., SI from objective perspective as explained in Section II-A. The goal-oriented SC systems explicitly define goals of communication tasks and filter out SI irrelevant to the task goals, i.e., SI from perceptual perspective as explained in Section II-A. Semantic-aware SC enables each communication agent to autonomously perceive and extract SI for communication tasks. In semantic-aware SC systems, SI is the implicit expression of individual observations or actions aimed at improving the efficiency of swarm collaboration.
- *In terms of the use of SI,* semantic-oriented SC systems emphasize recover the original data using the received SI. Goal-oriented SC systems focus on achieving task goals based on the received SI, such as object recognition in images and speech recognition. Semantic-aware SC systems establish connections between agents to share SI so as to enhance information synergy among agents and cooperatively complete tasks.
- *In terms of the level of communications,* semantic-oriented SC systems are designed to enhance the quality of SI extraction and transmission, which is the second level problem (i.e., the semantic problem) of communications. Goal-oriented SC systems and semantic-aware SC systems are designed to efficiently achieve task goals by transmitting necessary SI, which is the third level problem (i.e., the effectiveness problem) of communications.

We next introduce other classification methods for existing works on SC. According to fundamental theories, the existing works related to SC can be divided into semantic information theory-based SC, universal SC [30], [31], [32], [33], and AI based SC. According to modalities of source data, the existing works can be divided into SC for texts, SC for voices, SC for images, SC for videos, and SC for multimodal data. According to architectures of SC systems, the existing works can be divided into point-to-point SC system and SC networks.

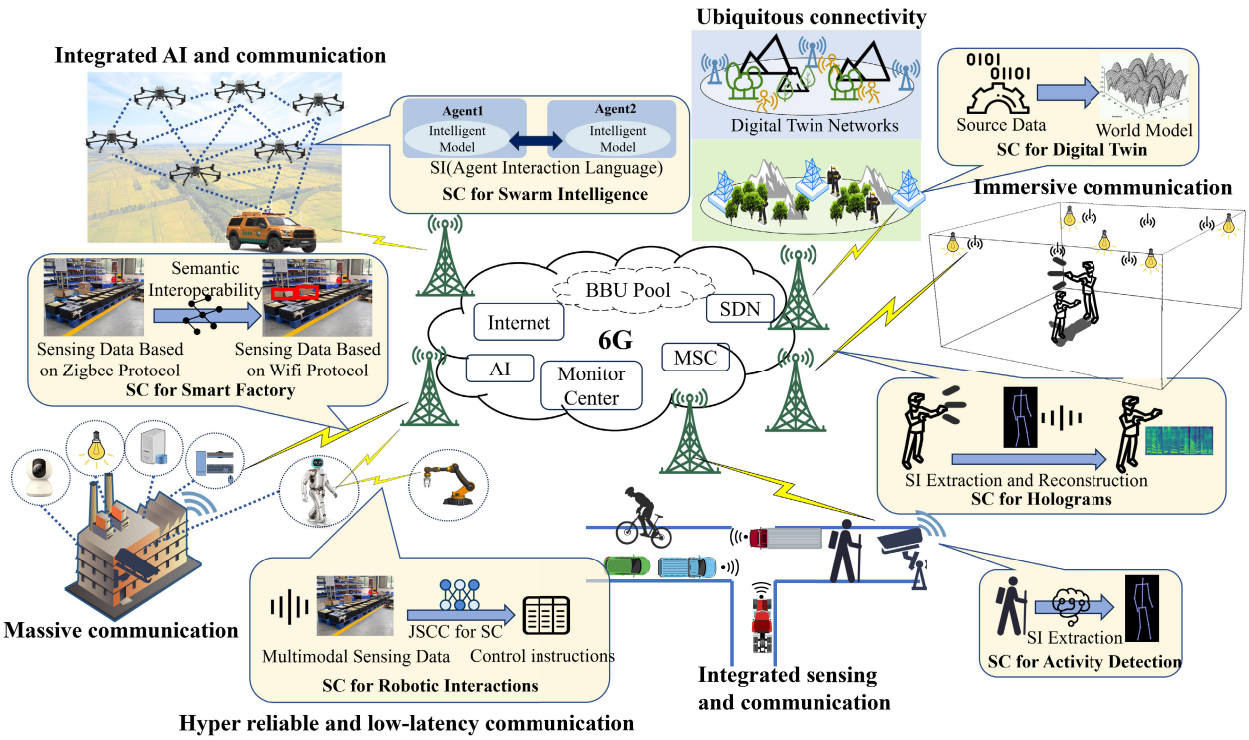

FIGURE 3. SC empowered 6G networks.

III. 6G WITH SC: MOTIVATING APPLICATIONS

The IMT-2030 (6G) proposes six major scenarios, as we specified in Section I-A. SC may provide promising solutions in each scenario. The typical use cases in each scenario are shown in Fig. 3. Meanwhile, the capabilities (e.g., edge intelligence) of future 6G networks may support the implementation and deployment of SC systems. Next, we will introduce the vision of mutual support between SCs and 6G networks.

A. SC EMPOWERED 6G NETWORKS

1) INTEGRATED AI AND COMMUNICATION

The scenario of integrated AI and communication is designed to support applications driven by AI and distributed computing, such as autonomous driving and autonomous collaboration between intelligent agents. The use of SC for swarm intelligence is shown in Fig. 3. SC technologies (especially semantic-aware SCs) promote the formation of machine interaction languages for efficiently achieving task goals, as described in Section II-C. Using SC in such scenario is expected to reduce the overhead of information exchange and understanding, improve the efficiency of agent collaboration, and alleviate the cognitive burden of human.

2) UBIQUITOUS CONNECTIVITY

The scenario of ubiquitous connectivity requires 6G networks to provide the ability to interconnect across domains and enhance network coverage. The typical use cases of this scenario include mobile broadband communication. For mobile broadband communication, the future 6G networks focus on the network coverage in rural, remote, and sparsely populated areas. SC is a promising technology for constructing digital twin networks of physical mobile networks based on AI model and knowledge model deployed in 6G network infrastructures. The use of SC for digital twin is shown in Fig. 3. Digital twin technologies white-box the network process and channel environments. Hence, based on SC, network operators can ensure the efficient deployment and normal operation of non-terrestrial communication systems and communication systems in harsh environments.

3) MASSIVE COMMUNICATION

The scenario of massive communication is an extension of IMT-2020 (5G) massive machine type communication (mMTC), aimed at supporting the access of ultra-large-scale machines such as devices and sensors. In this scenario, the machines requesting access are deployed for vertical applications such as smart cities, smart transportation, smart factory, and smart agriculture. The use of SC for smart factory is shown in Fig. 3. Since assess devices have different specifications and incompatible data formats, the core issue of massive communication lies in the interoperability of heterogeneous devices. SC technologies can model incompatible data into universal interpretable languages [34], such as knowledge graph (KG) and web ontology language (OWL), thus supporting semantic interoperability of heterogeneous

devices. Moreover, federated learning (FL) algorithms enable heterogeneous devices in SC systems to map heterogeneous and isolated data to a unified semantic space by joint training, thus providing a solution of interoperability of heterogeneous devices [35], [36], [37]. Hence, SCs achieve a significant reduction in wireless resource requirements while ensuring the age of information (AoI) of data. As a result, SC empowered 6G networks can further expand the scale of access machines [38].

4) INTEGRATED SENSING AND COMMUNICATION

The scenario of integrated sensing and communication requires 6G networks to provide spatial information of unconnected objects based on wide area multi-dimensional sensing, as well as the motion and surroundings of connected devices. The typical use cases of this scenario include activity detection, movement tracking, 6G assisted navigation, and providing sensing information for the extended reality (XR). The use of SC for activity detection is shown in Fig. 3. On the one hand, SCs can reduce the occupation of wireless resources by redundant information in perception data through SI extraction, and hence improving communication efficiency. On the other hand, SC technologies can achieve the trade-off between perception and communication capabilities of devices while meeting the requirements of 6G networks for device motion and spatial information. This means that the SI transmitted by SC systems can assist devices to improve the perception accuracy and perception range, while the perception ability of devices can reduce the requirements of information transmitted by SC systems.

5) IMMERSIVE COMMUNICATION

The scenario of immersive communication is an extension of IMT-2020 (5G) enhanced mobile broadband (eMBB), which requires 6G networks to transmit video, audio, and environmental data in a time-synchronized manner. The typical use cases of this scenario include immersive XR, remote multi-sensory telepresence, and holographic communications. The use of SC for holograms is shown in Fig. 3. SC systems support the receiver to maintain digital models of the transmitted objects, thereby reducing the data requirements to ensure the quality of experience (QoE) of immersive communications. For example, a speaker represented as a hologram is delivering a talk at a remote side. The receiver could maintain the timbre model and posture model of the speaker. Exploiting such models, the sender only needs to send SI of some of the multiple views necessary to reconstruct the hologram of the speaker at the receiver. Hence, SCs have significant advantages in reducing data rate requirements of immersive communication. In addition, in applications such as immersive XR and remote multi-sensory telepresence, SC systems support empathic and haptic communications, affective computing, and collaboration between different cyber-physical-spaces (CPS). Hence, the SC is a promising solution of achieving the seamless connection and intertwining of different kinds of intelligence, both natural and artificial.

6) HYPER RELIABLE AND LOW-LATENCY COMMUNICATION

The scenario of hyper reliable and low-latency communication is an extension of IMT-2020 (5G) ultra reliable and low latency communications (URLLC). The typical use cases of this scenario include robotic interactions as well as industrial environment for full automation, control, and operation. The use of SC for robotic interactions is shown in Fig. 3. In terms of reducing the latency, SC systems can reduce the redundant information in original data based on the shared KBs. Hence, compared with traditional communication systems, SC systems can effectively reduce transmission latency under the same channel state. In terms of improving the reliability, SC systems support end-to-end cross layer optimizations, as done in [39]. Simulation results show that, compared with traditional joint source-channel coding (JSCC) systems, the JSCC SC system can improve the peak signal-to-noise ratio (PSNR) by 10dB under low signal-to-noise ratio (SNR) conditions, as shown in Fig. 4.

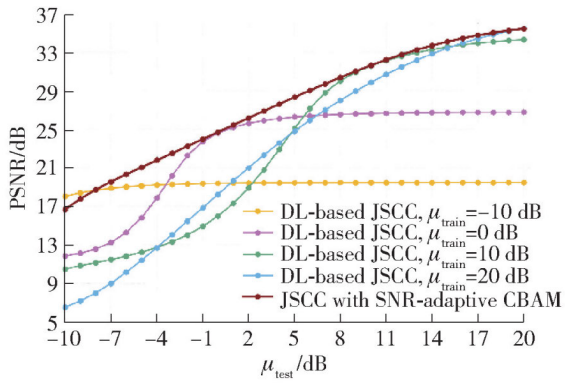

FIGURE 4. PSNR performance of JSCC SC system [39].

B. 6G NETWORKS EMPOWERED SC

The future 6G networks will integrate sensing, computing, and AI capabilities into networks, and, hence becoming a new type of digital infrastructure. Therefore, in addition to traditional communication services, 6G networks will provide a series of new capabilities related to AI and computing, such as distributed AI model training, AI model sharing, and distributed inference across international mobile telecommunications (IMT). As a result, 6G networks will act as a distributed neural system to support SC to build a unified prior KB for multiple communication agents based on distributed datasets. Moreover, 6G networks will support SC systems to dynamically update the KB based on the interaction SI so as to form a task-oriented interaction context among communication agents.

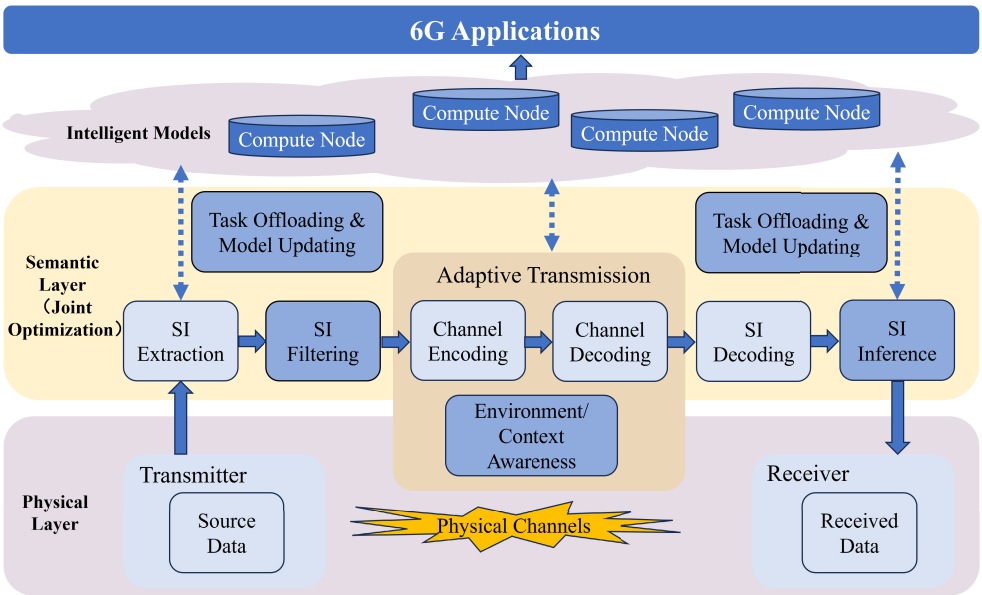

FIGURE 5. Architecture of SC systems.

IV. RESEARCH DIRECTIONS, CHALLENGES, AND STATE-OF-THE-ART

Recently, a number of existing works focus on SC technologies due to the advantages of deploying SC in 6G networks. Next, we first introduce the fundamental theories related to SC. Then, we provide a comprehensive overview of the key technologies of SC, including architecture of SC systems, SI extraction, SI transmission, and performance evaluation for SC systems.

A. FUNDAMENTAL THEORIES OF SC

1) SEMANTIC INFORMATION THEORY

Shannon's information theory [5] studied the first-level technical problems of communications based on the statistical theory of signals (the carrier of syntactic information). However, Shannon's information theory paid no attention to semantic information and pragmatic information [40]. The fundamental problems of SC can be obtained by extending Shannon's information theory. In particular, Shannon's information theory used the probability distribution of codewords (i.e., uncertainty) to calculate the information amount. As the uncertainty of information increases, the information entropy increases and so does the amount of information. Similarly, the amount of SI can be calculated by measuring the probability distribution of SI symbols. Here, SI symbols can be considered as the smallest unit of SI. Bar-Hillel and Carnap's semantic information theory (SIT) represented SI as unary predicate logic propositions [41]. Hintikka extended the domain of SIT from unary predicate logic to general predicate logic [42]. The SIT used logical probability of propositional statements to calculate the amount of SI. AS the uncertainty of SI increases, the entropy of the SI increases and so does the amount of SI [16].

2) STRONG SEMANTIC INFORMATION THEORY

For SIT, the amount of SI encoded by a proposition is inversely proportional to the likelihood of the truth of that proposition [43]. Hence, the amount of SI for the truth is small, while the amount of SI for unconventional events with small probabilities is large. As a result, many false propositions and incorrect sentences having a large amount of SI. To solve the problems, Floridi proposed the strong semantic information theory (SSIT) that defined a metric of truth value to represent the degree to which a proposition corresponds to the reality [44].

3) GOAL-ORIENTED COMMUNICATION THEORY

To answer the question of whether goals can be achieved through communication under conditions where communicating agents lack a common language or protocol, Juba first introduced the concept of universal semantic communication in [31] and [32]. Later, Juba established and developed the goal-oriented semantic communication theory in his works [30], [31], [32], [33]. The proposed theory studied a communication framework with two agents using different languages (or protocols) and a trusted third agent with the knowledge of the two languages (or protocols). The works in [30], [31], [32], and [33] took a variant of transmission control protocol (TCP) as an example to verify the feasibility of goal-oriented universal semantic communication between two agents using different protocols. In [30], [31], [32], and [33], Juba also quantified the relationship between protocol incompatibility and the complexity of achieving goal oriented universal semantic communication. The proposed goal-oriented semantic communication theory emphasized that the communication itself is not the goal, but a method for all agents to achieve an overall goal.

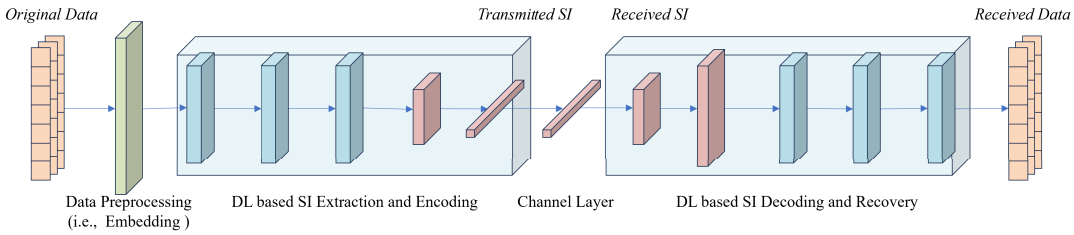

FIGURE 6. DL based SC systems [19].

B. SC SYSTEM ARCHITECTURE

Semantic communication systems can be implemented by extending traditional wireless communication systems. In particular, the intelligent models for SI extraction and parsing can be integrated into the coding and decoding modules, respectively, to enable SC systems to understand the source data and its SI [20].

The existing works [18], [21], [22], [45], [46], [47], [48], [49] introduced the components of the proposed SC systems. The work in [45] considered intelligent tasks as inputs for SI extraction in SC systems. In [46], the authors proposed an inference based implicit semantic perception communication network, which automatically infers complex implicit SI based on the received limited clues in the semantic decoding module. In [21] and [22], SI filtering based on SI importance was considered in the semantic encoding module. The authors in [18] considered semantic noise in the SC systems, including semantic extraction and encoding errors, packet errors caused by noisy channels, and semantic decoding errors. In addition, the work in [18] also introduced a perception module to sense the environment and context information, so as to enable the transmitter and receiver to locate SI in their local KBs. In [47], [48], and [49], the authors developed an adaptive semantic encoding module that can adapt to dynamic physical channel signal-to-noise ratio (SNR), and, hence enhancing the generalization of SC systems for different channel conditions.

The SC systems proposed in existing works can typically implemented by two approaches that are SI extraction independent of source-channel coding and SI extraction integrated with joint source-channel coding. The existing works such as in [22] and [50] used AI algorithms to extract SI, and then used traditional communication systems to transmit extracted SI. This implementation approach has two advantages. First, SI is interpretable and hence, one can easily validate the reliability of the SC systems. Second, the SI extraction module and SI transmission module are decoupled. Hence, the SI extraction module can be easily deployed in existing communication systems. However, this implementation approach cannot adjust the amount and the content of the extracted SI according to the channel environment. The existing works such as in [23], [51], and [52] integrated SI extraction module with joint source-channel coding communication systems and optimized the SC system performance by end-to-end training. The SC system implemented by such approach can obtain different SI coding schemes based on different training physical channels. However, the SI extraction module and SI transmission module are tightly coupled. The adaptability of SC systems to dynamic channels is analyzed in detail in Section IV-D.

Based on the existing works, we present a SC architecture, as shown in Fig. 5. In the proposed SC architecture, the transmitter extracts and filters the SI of the source data according to the requirements of 6G applications. The receiver decodes and infers the received SI so as to obtain the received data. The modules of SI extraction, SI filtering, channel encoding, channel decoding, and SI decoding can be jointly optimized according to the perceived environmental and contextual information. During the SC process, the BSs (or cloud centers) of 6G networks offload tasks to the transmitter and the receiver, and the transmitter and the receiver update the intelligent models deployed in the BSs (or cloud centers). Here, we note that modules such as task offloading and model updating, SI filtering, environmental and context awareness as well as SI inference can be optionally developed.

C. SI EXTRACTION

Semantic information extraction is the core module of the SC systems, which directly affects the applications that the systems can support, the implementation of other coupled modules, and the system performance. Based on different SI extraction algorithms, SI can be represented as three types of data structures. First, the SI extracted by DL algorithms can be modeled as vectors that represent the features of the source data in semantic spaces. Second, the SI extracted by KG construction algorithms [53] can be modeled as KGs that represent the entities and their relationships in source data. Third, the SI can be modeled as a hierarchical semantic tree (HST) that represents the hierarchical relationships between entities in source data.

1) DL BASED SI EXTRACTION

Deep learning algorithms can map images, speeches, texts, and multimodal data into semantic spaces. Since training DL models require explicitly defined loss functions, the DL based SI extraction models are trained for specific tasks. An example architecture of DL based SC system [19] is

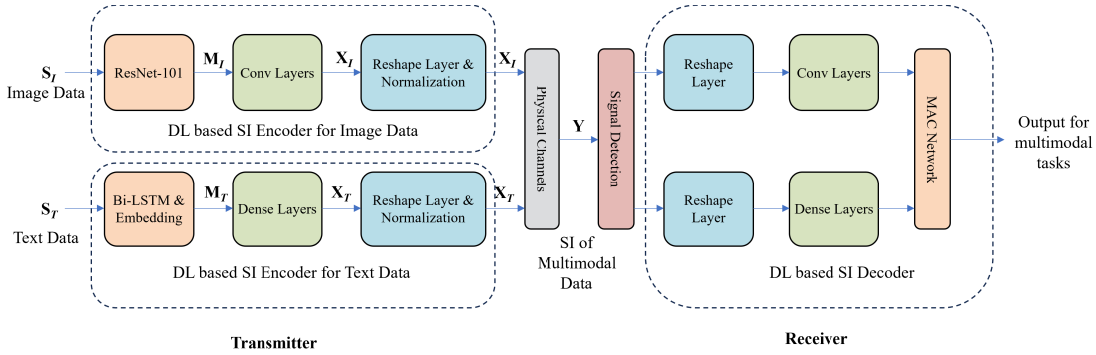

FIGURE 7. A SC system for multimodal data transmission [54].

shown in Fig.6. Next, we introduce the state-of-art works on DL based SI extraction for each type of data from the task perspective.

- *SI extracted from image data*: DL based image SC systems can be used for a large number of tasks such as image recovery, image classification, semantic segmentation, and object detection. For the image recovery task, the authors in [23] and [51] used a convolutional auto-encoder (CAE) to extracted feature vectors as SI from image data, and used a convolutional decoder at the receiver to restore original image based on the received SI. The authors in [52] presented the generative adversarial networks (GANs)-based image semantic coding whose base layer preserves SI of images and enhancement layer restores the fine details of images. The work in [55] used convolutional neural networks (CNNs) to extract the categories, spatial arrangements, and visual features from images as SI, and fused local and global SI by attention modules for image reconstruction. In [56], the authors proposed a dual-path semantic encoder and decoder that extract and reconstruct SI at both information and concept levels and are optimized in a neural network (NN) based end-to-end SC system. The image SC system proposed in [57], inspired by the way humans understand and convey images, modeled SI of images as their text descriptions and reconstructed the original images using text-to-image generative algorithms.
  For the image classification task, the authors in [24] developed an image SC system based on residual networks (ResNets) that can fuse SI extracted from multiple original images. In [58], the authors used adversarial training to effectively improve the robustness of deep auto-encoder (DAE) based SI extraction. The authors in [45] eliminated redundant SI represented by feature vectors while ensuring image classification accuracy. The work in [59] divided original images into semantic blocks and trained a deep reinforcement learning (DRL) model whose goal is to maximize image classification accuracy by selecting semantic blocks.
  The authors in [26] and [60] designed image SC systems for semantic segmentation tasks. The designed SC systems extracted the category and boundaries of the target object from a image as SI, and the receiver reconstructed a mask image with the object category after receiving SI. Moreover, the works in [61] and [25] proposed a DL based JSCC SC system that can support multiple tasks such as object detection, image classification, and image retrieval. The differences and advantages of existing DL based methods for extracting SI from image data are summarized in Table 2.
- *SI extracted from speech data*: DL based speech SC systems can be used for tasks such as speech recovery [62], [63], [64] and speech to text transcription [65], [66], [67]. For the speech recovery task, the authors in [62] proposed a SC system named DeepSC-S that used squeeze-and-excitation (SE) network and attention mechanism to identify and recover critical speech information. The work in [63] used two kinds of NNs to extract SI features and acoustic features from speech signals, and fused two kinds of features to avoid speech distortion. In [64], the authors used deep neural networks (DNNs) to extract feature vectors of original speeches as SI, and established an entropy model to estimate the importance of feature vectors.
  For the speech to text transcription task, the authors in [65] used NNs to extract SI from speech signals and the SI was decoded into text with semantic fidelity. Based on the work [62] and [65], the work in [66] designed a SC system for speech synthesis tasks. In particular, the transmitter converted the speech signal into text, and then the receiver synthetized speech based on the received text using local acoustic KBs. In [67], the authors proposed a SC system that can support speech to text transcription task and speech recovery task simultaneously. The differences and advantages of existing DL based methods for extracting SI from speech data are summarized in Table 3.
- *SI extracted from textual data*: DL based text SC systems can be used for tasks such as text recovery [68], [69],

TABLE 2. A summary of works on DL based SI extraction for image data.

Source Data	Task	AI Model	SI Representation	Works	Advatages
Image Data	Image Recovery	CAE	Feature Maps	[23], [51]	Simple and symmetrical DL model structure. Supporting JSCC.
		GAN/CNN	Hierarchical SI, e.g., pixel-level, feature-level, and concept-level SI.	[52], [55], [56]	Conforming to the nucleus of human thinking, which is "knowledge extraction and abstraction". [15].
		Generative Models	Text descriptions of images.	[57]	SI is interpretable.
	Image Classification	CAE/DAE/ ResNets	Feature Maps	[24], [45], [58]	Simple and symmetrical DL model structure. Supporting JSCC.
		DRL	Image Blocks	[59]	SI is interpretable.
	Semantic Segmentation	GAN	Category & Boundaries	[26], [60]	SI is interpretable.
	Multiple Tasks	DNN	Hierarchical Feature Maps	[25], [61]	Conforming to the nucleus of human thinking, which is "knowledge extraction and abstraction". [15].

TABLE 3. A summary of works on DL based SI extraction for speech data.

Source Data	Task	AI Model	SI Representation	Works	Advatages
Speech Data	Speech Recovery	DNN/SE	Semantic & Acoustic Feature Maps	[62]–[64]	Simultaneously supporting semantic recovery and perceptual recovery.
		DNN	Text	[66]	SI is interpretable.
	Speech to Text	DNN	Feature Maps	[65]	Simple and symmetrical structure. Supporting JSCC.
	Multiple Tasks	DNN	Semantic & Acoustic Feature Maps	[67]	Simultaneously supporting semantic recovery and perceptual recovery.

[70] and speech to text summarization [71]. Different from images and speeches, textual data cannot be directly represented as numerical vectors or matrices that the learning model can process. Hence, before SI extraction, it is necessary to embed the texts into numerical vectors or matrices based on language models. Word embedding technologies [72] maps words from a vocabulary to a continuous vector space, giving language a representation of real numbers. Recently, the large language model (LLM) has attracted a lot of attention, and is able to obtained effective language representation. In the DL based text SC systems, the transmitter first use language models to represent original texts, and then further extracts the task oriented SI using DL algorithms. For the text recovery task, the authors in [19] proposed a DNNs based text SC system. Based on [19], they transformed the text SC system to be lightweight and distributed and deployed in IoT networks in [68]. The work in [69] used Transformer algorithm to extract SI from original texts and aimed to maximize the sentence similarity between the recovered texts and the original texts. In [74], the authors integrated an adaptive computation time (ACT) module into an end-to-end text SC system, so as to enable the SC system to adaptively adjust training rounds according to the SI. The authors in [70] integrated self attention mechanism and adversarial training with the DNNs based SI extraction module that can resist semantic noise and physical noise.

For the text summarization task, the authors in [71] proposed a text SC system in which the base station (BS) extracts the abstracts of long texts using long short-term memory (LSTM) algorithms and transmits the abstracts to users to realize the efficient transmission of long texts. The work in [75] proposed a multi-task oriented text SC system, which used the bidirectional encoder representation (BERT) model to extract the SI from original texts and parse the received SI for multiple tasks. The differences and advantages of existing DL based methods for extracting SI from textual data are summarized in Table 4.

- *SI extracted from multimodal data*: DL algorithms enables SC systems to map the SI of multimodal data into a unified semantic space and obtain the vector representations of SI. Hence, when multimodal data provides complementary or corroborative information for a specific task, the SI of multimodal data can be fused in the process of SI extraction, so as to improve the performance of task-oriented SC systems. As shown in Fig.7, the authors in [54] proposed a task oriented multi-user SC system in which some users transmitted images and others transmitted text to query information about the images. The proposed SC system obtained the correlation of SI of multimodal data by jointly training SI extraction models for image and text data. In [76], the authors extracted the SI from multimodal data for the tasks of image retrieval, machine translation, and visual question answering. Different from the works such as

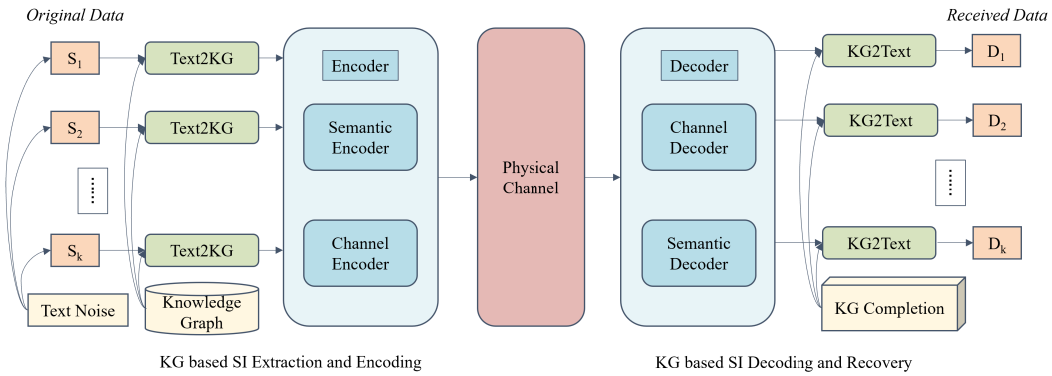

FIGURE 8. A KG based SC system [73].

TABLE 4. A summary of works on DL based SI extraction for texual data.

Source Data	Task	AI Model	SI Representation	Works	Advatages
Textual Data	Text Recovery	DNN/ Transformer	Feature Maps	[69], [70]	Simple and symmetrical structure. Supporting JSCC.
		DNN	Word-level & Sentence-level Feature Maps	[19], [68], [74]	Conforming to the nucleus of human thinking, which is "knowledge extraction and abstraction".
	Text Summarization	LSTM	Feature Maps	[71]	Suitable for long text transmission.
	Multiple Tasks	BERT	Feature Maps	[75]	Simultaneously supporting multiple tasks.

in [54] and [76] that use DL algorithms to extract SI, the authors in [77] proposed to use binary hash codes to encode the SI of texts, images, and videos. The system proposed in [77] used NNs to map the hash code to the different semantic space according to the data type and recovered the original data based on the hash function at the receiver. The differences and advantages of existing DL based methods for extracting SI from multimodal data are summarized in Table 5.

2) KG BASED SI EXTRACTION

The knowledge graph is a kind of data structure that describe the entities (i.e., objects or concepts) and their relationships using graphs [78]. The nodes in the KG represent the entities and the edges represent the relationships between entities. The basic unit of a KG is the triple formed as "*entity-relationship-entity*" or "*entity-attribute-value*". The scene graph (SG) is a visual KG that describes visual relationships between entities, such as spatial positional relationships and motion trends [79]. By using intelligent algorithms such as named entity recognition (NER), relation extraction (RE), and knowledge fusion, the SI of all types of original data can be modeled as KGs, and then serialized into sequences of semantic triples for transmission and inference. For example, in hyper reliable and low-latency communication scenarios of 6G networks, the intelligent agents model the SI of sensor data by KGs and perform inference based on the local KB so as to generate appropriate control instructions. An example of KG based SC system proposed in [73] is shown in Fig.8. Next, we introduce the state-of-art works on KG based SI extraction for each type of textual and image data.

- *SI extracted from textual data*: The work in [21] used an information extraction system to extract semantic triples from texts and modeled as KGs, and the receiver used an graph-to-text generative algorithm to recover the original texts based on the received triples. The authors in [80] proposed a KG alignment algorithm that can locate the SI of the source text in the KB and performing inference, thus effectively correcting the SI of original texts. Different from the works in [21] and [80] that directly transmits the semantic triples, the work in [73] encoded the semantic triples in SI of the texts into the indexes of triples in the shared KB, and, hence further reducing the size of data that needs to be transmitted. Similarly, in order to reduce the size of transmitted data, the authors in [81] only transmitted the keywords in the extracted SI, and the receiver recovered the original text based on the received keywords and shared KB. In [82], prior knowledge was used to improve the decoding performance of DL based SC systems. In particular, the transmitter mapped the local KB and the original text to the same semantic space, and the receiver jointly used the received SI and the corresponding knowledge to recover the text.
- *SI extracted from image data*: The authors in [22] and [50] used object detection and RE algorithms to extract SG from images as their SI. In [83], the

TABLE 5. A summary of works on DL based SI extraction for multimodal data.

Source Data	Task	AI Model	SI Representation	Works	Advatages
Multi-modal Data	VQA	Transformer	Feature Maps	[54]	Supporting feature level fusion for SI of multimodal data.
	Multiple Tasks	DNN	Feature Maps	[76]	
	Image recovery	DNN	Binary Hash Codes	[77]	Low computational overhead.

authors proposed an image SC system in which the receiver supplemented the received SG using inference algorithms and reconstructed the original image based on the supplemented SG. The work in [84] extracted 3D SGs from the environment information perceived by robot cluster as SI and compressed 3D SGs using graph theory to further reduce communication overhead. In order to preserve the structural information of original images, the work in [85] used a graph convolutional neural network (GCN) to extract the sizes and locations of the entities in original images. To improve the quality of reconstructed images, the authors in [86] proposed a semantic reconstruction model for KG based image SC systems, which can generate reasonable geometric layouts based on the visual relations between entities and generate entities based on the their shapes and appearances.

3) HST BASED SI EXTRACTION

The hierarchical semantic tree (HST) organizes the entities into a tree structure. HSTs can be divided into two types according to the relationships between entities, which are ontology-based HSTs that only contains strict "is-a" relationships, and taxonomy-based HSTs that contains hypernym-hyponym relationships. The prerequisite for SC agents to be able to communicate based on the SI represented by HSTs is to define consistent meta SI (i.e., the leaf nodes of the HSTs). The definition of meta SI in HSTs should satisfy five conditions that are finiteness, Comprehensiveness, independence, non-circularity, and primitivity [87].

The existing works in [15], [88], and [89] studied the HST based SI extraction. The authors in [15] proposed a HST based SC systems that represented the SI of messages by a vector composed of n-dimensional meta SI. In [88], the authors provided an example of meta SI, which is the basic human senses including sight, hearing, smell, taste, and touch. In [89], the authors defined the meta SI as the "semantic base" (Seb) for SC networks, and explained that the SI of multimodal data can be ultimately represented by Seb sequences via multi-level transformation. The works in [89] indicated that using sequences of meta SI to represent the SI of original data enables SC systems to reconstruct information in a highly integrated manner.

D. SI TRANSMISSION

Next, we introduce the semantics-oriented transmission technologies, including SI filtering, adaptive SI transmission, resource allocation for SC networks, and multiple access techniques for SC networks.

1) SI FILTERING

A number of existing works studied the SI selective transmission strategies to improve the efficiency of SI transmission. For example, in immersive communication scenarios of 6G networks, the receiver maintains digital models of the transmitted objects, and, hence, the transmitter can filter out the SI related to the object model. The works in [21], [22], and [25] investigated the SI filtering schemes for semantic-oriented SC systems. The authors in [21] and [22] respectively used attention mechanisms and cosine distance to evaluate the relevance between semantic triples and original data, and then transmit the most relevant triples using limited wireless resources. In [25], the authors used a gradient weighted class activation mapping (Grad CAM) model to calculate the importance of feature vectors in the image SI extracted by CNNs, and visualized the feature importance as heatmaps to guide SI selection in SC systems.

The works in [55], [90], [91], and [92] investigated the SI filtering schemes for goal-oriented SC systems. In [55], the authors proposed a RL-based semantic bit allocation model that can preserve and reconstruct task-related SI and discard less important SI. The authors in [90] used neural symbolic (NS) to model the SI of images and selected image SI according to the information requirements of receivers and the security requirements of SI. The work in [91] used domain adaptation techniques to select the SI extracted from multimodal data for specific tasks, and, hence effectively reducing the amount of transmitted SI. However, the existing works, such as [55], [90], and [91], assumed that the transmitter has the knowledge of the tasks or requirements of receivers. The authors in [92] proposed a transfer learning based SC system that enables the transmitter without the knowledge of the tasks to effectively select SI to satisfy the requirements of receivers.

2) ADAPTIVE SI TRANSMISSION

A SC system using DL algorithms can effective work only if the transmitted data follows the distribution of the training data. In particular, deploying the trained JSCC based SC systems requires the actual channel state is consistent with the channel state during training. Hence, a challenge of deploying SC systems is adaptive SI transmission. The works in [47] and [64] input both SNRs of transmission channels and original data into SI extraction models, and, hence, the trained SI extraction models can dynamically adjust the compression

ratio and channel code rate of original data. The authors in [63] and [93] integrated a SNR adaptive layer into the DNN based SI extraction models to achieve adaptive compression and channel coding of the extracted SI. In [48], a spatial modulation layer for processing channel state information (CSI) is added after each convolutional layer of a DL based image SC system to adaptively compress the SI of images.

Different from the above works of using adaptive layers to achieve adaptive SI transmission, the work in [49] introduced a differentiable policy network that enables SC system to adjust the SI quality to adapt to different channel conditions, image content, and transmission rates. In [94], the authors proposed a variable length SI coding scheme that can adjust the number of activated neurons in DNN based SC systems. The simulation results shows that the proposed SC systems overcame the impact of harsh channel environments by increasing the dimension of encoded SI (i.e., feature vectors). The authors in [95] used nonlinear transform source-channel coding and an SI entropy model as the prior for decoding, so that the SC system can adapt to different distributions of original data. Based the results in [95], the authors in [96] further proposed an online learning based JSCC SC system that can effectively adapt to dynamic original data and SNR by updating the pre-trained model online.

3) RESOURCE ALLOCATION FOR SC NETWORKS

A large number of existing works studied resource allocation (RA) technologies for traditional wireless communication systems. The existing works such as in [97], [98], and [99] allocated wireless resources based on the transmission requirements (e.g., priority and latency) of the complete original data and, hence, each bit in the original source data is treated equally. However, such RA technologies cannot allocate wireless resources based on the characteristics of SI contained in each part of the original data, such as the amount and importance of SI. To improve the efficiency of SC systems, one of the crucial challenges is the semantic oriented RA.

The works in [100], [101], [102], and [103] studied the RA technologies for DL-based SC systems. The authors in [100] used the deep deterministic policy gradient (DDPG) algorithm that prioritized data containing more SI to occupy limited transmission power and bandwidth, thus maximizing the long-term transmission efficiency. In [101], the authors allocated wireless resources based on the confidence of SI in image data so as to improve the quality of important SI transmitted by SC networks. The work in [102] iteratively optimized the compression rate and RA for SI according to the correlation between SI and the task goal to maximize the probability of task success. The authors in [103] defined the semantic spectral efficiency that is used to optimize the channel assignment and the number of transmitted semantic symbols.

The works in [21], [22], and [50] studied the RA technologies for KG-based SC systems. The authors in [50] derived an accurate expression for the packet loss rate of semantic triplets under dynamic wireless fading channels, and designed a multi-user power allocation scheme based on game theory. In [21] and [22], the authors used RL algorithms to allocate subcarrier based on the importance distribution of semantic triplets extracted from texts and images, respectively, to improve the accuracy of text recovery and image classification.

4) MULTIPLE ACCESS FOR SC NETWORKS

The authors in [104] proposed the model division multiple access (MDMA) technology by excavating the shared and personalized SI for multiple users from the high-dimensional semantic space. In particular, MDMA enables the SC systems to transmit shared SI to multiple users over a shared frequency band and transmit personalized SI to each user over a private frequency band. The receiver integrates the shared and personalized SI for data recovery. This work analyzed the performance gain of MDMA compared to traditional multiple access technology from the perspective of information theory, and verified the performance of MDMA in the additive white Gaussian noise (AWGN) channel under the low SNR condition. Based on the MDMA technology, the authors in [105] proposed the concept of semantic orthogonal signals and an Orthogonal-Model Division Multiple Access (O-MDMA) technology. The proposed O-MDMA migrated the anti-interference capability of Deep JSCC to the multi-user capacity and can be applied to any semantic model.

E. SC PERFORMANCE EVALUATION

The metrics that used to evaluate the performance of SC systems mainly include semantic channel capacities, task oriented metrics, semantic perception metrics, and metrics for SC systems. Next, we show each type of metrics for SC systems.

1) SEMANTIC CHANNEL CAPACITY

As a natural extension of Shannon's information theory, the authors in [106] proposed a measurement approach for SI represented by propositional statements by integrating fuzzy set theory and logical probability. In [18], the authors derived the semantic channel capacity based on the mutual information between semantic symbols and expected meanings. However, the measurement approaches proposed in [18] and [106] are based on logical probability and can only be used to the SC systems that represent the SI by propositional statements. Moreover, the works in [18] and [106] faces challenges in practical applications as they ignored the impact of wireless channel state on the performance of SC systems. The authors in [103] defined the semantic spectral efficiency based on the information amount of each single semantic symbol. To compare the performance of SC systems with traditional communication systems, the authors in [103] studied a conversion algorithm between traditional spectral efficiency and semantic spectral efficiency.

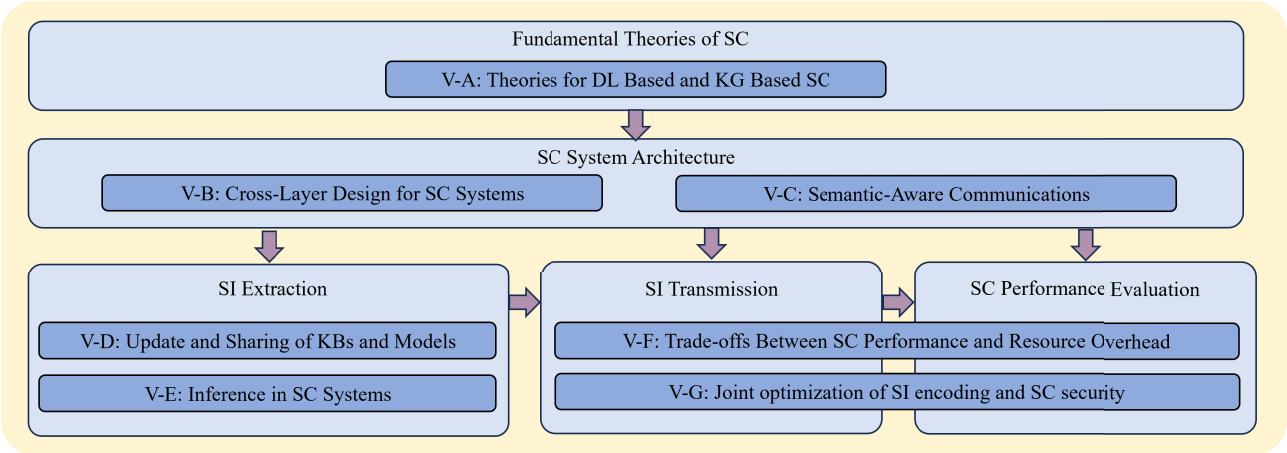

FIGURE 9. Open issues of SC technologies.

2) TASK ORIENTED METRICS

Numerous existing works introduced the metrics defined for intelligent tasks in the fields of natural language processing (NLP) and computer vision (CV).

- For text SC systems, the metric of cosine distance [19], bilingual evaluation understudy (BLEU) [107], and metric for evaluation of translation with explicit ordering (METEOR) [108] are commonly used. The cosine distance calculates the distance in semantic space between the vectorized words in original texts and the vectorized words in received (recovered) texts. However, mapping words in the semantic space can disrupt the word order of the texts. Hence, the metric of cosine distance cannot recognize the semantic errors caused by mess word orders. The metric of BLEU and METEOR evaluate the similarity of sentences by counting the number of ordered matching words and are fault-tolerant to synonyms in two texts based on synonym substitution. However, While BLEU and METEOR are able to evaluate the accuracy of the SI in received (recovered) texts, they cannot evaluate the completeness of the received SI. The works in [21] defined a semantic similarity metric for text SC systems, which jointly considered the semantic accuracy and completeness of the recovered texts.
- For image SC systems, the metric of image classification accuracy [109], object detection F1 score [110], and Multiple Object Tracking Accuracy (MOTA) [111] are commonly used for image classification tasks. Moreover, the authors in [19] and [68] evaluated the performance of an image recovery task oriented SC system by the pixel error rate between the original image and the recovered image. The works in [54], [76], and [90] deployed a visual question answering (VQA) model on the receiver to evaluated the accuracy and completeness of the received SI according to the accuracy of answers. In [22] and [112], the authors studied the cross-modal evaluation methods for image SC systems. In particular, the work in [22] calculated the semantic similarity between SGs and images based on the attention mechanism. In [112], the authors used the semantic proposition image caption evaluation (SPICE) algorithm to evaluate the semantic similarity between the original images and the received SI represented by texts.

3) SEMANTIC PERCEPTION METRICS

Semantic perception metrics are used for human involved SC systems, and emphasize the semantic and perceptual consistency between the original data and received data. The authors in [62] evaluated the performance of a voice SC system based on speech signal-to-distortion ration and perceptual evaluation of speech distortion. In [52], the authors used multiple metrics to train and evaluate an image SC system whose goal is to enable the recovered image to be visually pleasant and semantically consistent. The metrics considered in [52] include using Fréchet inception distance (FID) and kernel inception distance (KID) to calculate the distribution differences between original and recovered images, as well as using learned perceptual image patch similarity (LPIPS) and mean intersection over union (mIoU) to quantify the perceptual similarity of the recovered image.

4) METRICS OF SC SYSTEMS

The performance of SC systems can be evaluated from multiple dimensions, such as timeliness, energy efficiency, security, robustness, and adaptability.

- *Timeliness*: The works in [113] and [114] introduced the age of information (AoI) that describes the freshness of real-time data to evaluate the timeliness of the data transmitted by SC systems for real-time control systems.
- *Energy efficiency*: The authors in [115] calculated the computing energy consumption for SI extraction and

data recovery, as well as communication energy consumption for SI transmission in cellular SC networks.

- *Security*: Compared to traditional communication systems that directly transmit original data, SC systems transmit task-oriented SI based on the shared KBs, and, hence improving the system security. However, the construction of shared KBs faces new security risks. Moreover, although SC systems only transmits SI, partial private information can be reconstructed from feature vectors through model inversion and knowledge inference, thus resulting in private information leakage. In [116], the authors defined the semantic privacy metric and the semantic-aware utility metric based on information theory to evaluate the privacy leakage in location-based services. The works in [117] designed a loss function that can increase the uncertainty (i.e., entropy) of SI to enhance the security of SC systems.
- *Robustness*: DL based SC systems are susceptible to semantic noise and semantic attacks. The work in [14] evaluated the robustness of SC systems by the stability of the SC system over noisy data.
- *Adaptability*: The works in [63] and [93] used the quality of the data recovered based on the SI transmitted over dynamic channels to evaluate the adaptability of SC systems. The works in [97] evaluated the task adaptability of a meta learning based SC system by the number of iterations required for fine-tuning the SC system for different tasks.

V. OPEN ISSUES AND FUTURE OPPORTUNITIES

In the previous section, we have outlined the general research directions of SC technologies. Despite a considerable number of studies on SC technologies, there are still many key open problems that must be investigated, such as the theories, architectures, sustainability, efficiency, and security of SC systems. Next, we discuss the open issues and future opportunities involved in each research direction in SCs, as shown in Fig. 9.

A. THEORIES FOR DL BASED AND KG BASED SC

The exiting fundamental theories for SC are specified in Section IV-A. The SIT and SSIT are investigated for SC systems that model the SI as logic propositions. The goal-oriented communication theory provides a foundation for implementing cross protocol SC based on trusted third parties. However, none of the existing works studied the fundamental theories for DL based and KG based SC systems. For DL based SC systems, the design of neural network structures and the setting of hyperparameters in SC systems still lack theoretical basis. Moreover, the unexplainability and incomprehensibility of SI pose challenges to the reliability and maintainability of the DL based SC systems. For KG based SC systems, there is no theory to support the completeness measurement of KG based SI. Hence, the fundamental theories for DL Based and KG Based SC are bottlenecks required to be overcome so as to guarantee the performance of SC systems in practical applications. Since the SC technology integrates AI with communications, the theories of SC based on DL and KG can be traced back to the theories of AI and information theory. For example, the probability theory based optimization model, such as Bayesian optimization, can provide insights for the theoretical derivation of DL model design and optimization in SC systems.

B. CROSS-LAYER DESIGN FOR SC SYSTEMS

From the perspective of the network protocol stack, the SI extracted by SC systems can be encapsulated as application layer packets and transmitted following traditional network protocol stacks, such as the open systems interconnection (OSI) model [118] and the TCP/IP stack. However, traditional network protocol stacks reduce the complexity of network design by clarifying the functions of each layer and interaction interfaces between layers. Based on the exiting stacks, the nodes in traditional communication networks do not care about the meaning of the transmitted data. Hence, the information exchanged between layers and nodes can be regarded as homogeneous bit sequences. In SC systems, the SI extracted at the application layer is coupled with the physical layer and the network layer protocols, as well as dynamic network environments. Traditional network layered protocol stacks are difficult to effectively support SCs. Therefore, designing a SC network architecture that supports cross layer joint optimization is a fundamental issue that urgently needs to be addressed.

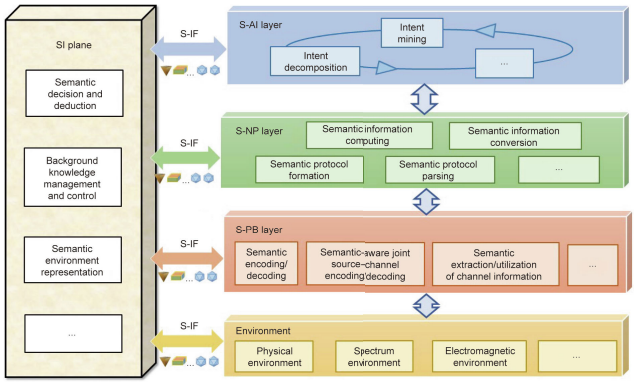

FIGURE 10. IE-SC network architecture [89].

The existing works such as in [51], [52], and [76] used the JSCC technology to train end-to-end SC systems so as to achieve automatic source channel encoding for SI. The authors in [27] jointly optimized the codebook selection in the physical layer and the SI extraction in the application layer to improve the efficiency of the Cart-Pole control system using SC technologies. The DL based SC system with the digital modulation module cannot be trained end-to-end using existing gradient descend algorithms, since the digital modulation module maps continuous feature vectors (i.e., the SI) into discrete constellation symbols. To this end, the

authors in [119] developed a joint coding-modulation scheme for digital semantic communications with binary phase shift keying (BPSK) modulation that map the continuous SI in the application layer to the likelihood of constellation points in the physical layer. JSCC is a widely studied cross-layer SC architecture. Future works can explore the joint optimization of JSCC and other elements (e.g., the waveform design) of SC systems.

In terms of the architecture design for SC networks, the authors in [89] simplified the classical OSI model and proposed an intelligent and efficient semantic communication (IE-SC) network architecture, as shown in Fig. 10. The proposed architecture developed three semantic-empowered layers, which are physical-bearing (S-PB) layer, network protocol (S-NP) layer, and application-intent (S-AI) layer. The above three layers and the physical environment are interacted using semantic information flows (S-IFs) on a vertical semantic intelligent plane. Despite a promising SC network architecture for future 6G network, the architecture proposed in [89] cannot be compatible with traditional network protocol stacks. An engineering solution is to achieve the coexistence of SC architecture and traditional communication architecture based on network function virtualization (NFV) or network slicing technologies. In addition, based on the IE-SC network architecture, there are several critical challenges that need to be researched, such as semantic oriented waveform design and modulation, semantic oriented routing protocol design, and semantic oriented RA.

C. SEMANTIC-AWARE COMMUNICATIONS

The existing works on SCs mainly focuses on semantic-oriented SCs and goal-oriented SCs. Due to the fact that mapping an intelligent system to the semantic space is a highly challenging problem, only a few works studied that intelligent agents utilize SC systems to automatically perceive and transmit task oriented SI. In [27], the authors considered an agent sending observation information to another agent using a SC system to guide it to optimize its behavior. In [28], a remotely connected agent transmitted the SI of the observations, and analyzed the trade-off between the accuracy of the SI representation of actions and the system reward. However, the works in [27] and [28] did not consider the cooperation among multiple agents. The authors in [29] proposed a cooperation control strategy for a multi-agent actor-critic based RL algorithm that used actor network to determine the action of each agent and used a critic network to evaluate the policies based on the SI of observations. However, none of the existing works considered the partial observability and communication limitations of agents in semantic-aware networks. Hence, a key issue in the semantic-aware networks is to infer and fuse the partial SI perceived by each agent to obtain the global perception information. A promising solution is to use multi-agent RL algorithms that enables the agent to locally build the relationship between the fused SI (e.g., the fused features) of the "observations" and the collaborative task performance.

D. UPDATE AND SHARING OF KGs AND MODELS

1) UPDATE AND SHARING OF KGs

In KG based SC systems, the effective SI extraction and parsing rely on the matched local KB stored at each agent (or a shared KB stored at cloud). In particular, the transmitter locates entities recognized from original data in the local KB to understand the original data, and models the SI of original data as semantic triples to transmit. Here, we note that the semantic triples extracted from the original data may not contains in the shared KB, which means that the original data contains the information that is not yet known to the agents. The receiver locates the received entities in the local KB to parse the received SI. As the number of communication rounds increases, communication agents need to continuously update their local KBs (or the shared KG) based on the received SI so as to achieve SCs that conform to the communication context.

The work in [120] studied a caching based KB update mechanism consisting of three steps that are kB initialization, device request subgraph query, and partial KB update. The goal of updating the KB is to maximize the caching KB value that builds the relationship between the caching KBs, communication delay, and data recovery accuracy. However, further exploration is required in terms of fusing incompletely matching KBs among communication agents. A KG alignment enabled SI decoder can be developed to solve this issue.

2) UPDATE OF INTELLIGENT MODELS

In DL based SC systems, the applications of trained SI extraction and parsing models rely on the consistent distributions of the data to be transmitted and the training data. For example, a DL based SC system trained on road traffic images cannot be directly applied to transmit satellite remote sensing images. Especially for the JSCC SC systems, the communication performance will greatly reduce if the distributions of the actual CSI and the training CSI are inconsistent. Hence, to ensure the availability of DL based SC systems, each agent must cooperatively retrain or fine tune the intelligent models according to new application data and communication environments.

To enable intelligent models to use learned knowledge to accelerate training on a small amount of new data, a series of DL frameworks were proposed, such as transfer learning, online learning, and lifelong learning. In [96], the authors proposed an online learning based JSCC SC system that updated the pre-trained model in a lightweight online manner so as to enable the SC system to adapt to specific use cases. The authors in [121] introduced a lifelong learning algorithm in a inference-supported SC system that enables the receiver to automatically update the inference model when it receives new semantic triples. However, the existing works such as in [96] and [121] required to retrain the models using limited local computing resources after the data distribution or the environment changes. Hence, the core concern is the

latency, computational overhead, and adaptability of using local data to update intelligent models. Distributed training strategies, such as FL, can jointly use the resources and data of multiple agents to train a robust SC system [122], [123], [124]. In particular, each agent first trains a SC system based on its own dataset. Then, each agent shares the parameters of its local SC system and aggregates a global SC system that can be widely used in different network applications and environments based on the parameters shared by other agents. As a result, FL algorithms can improve the adaptability and robustness of the DL based SC systems without sharing local data.

E. INFERENCE IN SC SYSTEMS

Most previous research on SC technologies focused on transmitting explicit SI, such as objects that can be directly identified from original data. However, implicit SI contains in original data is difficult to express, identify, or recover for SC systems. For example, in [125], the transmitter transmits a voice message being "What is a Tweety?". The entity "Tweety" can be interpreted in multiple ways, such as the name of a mobile app, a type of bird, or a character in a TV show. When the receiver parses the meaning of "Tweety", the implicit information, such as the context and the background of the transmitter, is required for inference.

The works in [121] and [125] provided two solutions. In [121], the authors used an embedding model to map the SI modeled as KGs into a low-dimensional semantic space, and designed an inference function used to infer implicit SI that cannot be directly extracted from origianl data, such as incomplete entities and missing relations. The authors in [125] proposed a generative imitation-based reasoning mechanism learning (G-RML) algorithm that enables the receiver to learn and imitate the implicit semantic reasoning process of transmitter. As a result, the receiver can obtain a consistent understanding of SI with the transmitter. The goal of inference in SC systems is to mine implicit SI in original data. To achieve this goal, spatial mapping based representation inference algorithms, such as TransG [126] can be used for DL based SC systems. KG completion algorithms, such as association rule mining under incomplete evidence (AMLE) [127]) can be used for KG based SC systems.

F. TRADE-OFFS BETWEEN SC PERFORMANCE AND RESOURCE OVERHEAD

In general, as the amount of the SI extracted from the original data increases, the granularity of SI decreases and the precision of SI increases. The fine-grained and precise SI enables the receiver to complete complex tasks but requires amounts of communication and computing resources. In addition, to ensure the quality of SI extraction and parsing, non-negligible computing and storage resources are required to update and share KBs and intelligent models, especially in SC networks with numerous participants. Therefore, it is crucial to achieve a trade-off between performance and resource overhead of SC systems.

1) JOINTLY SCHEDULE COMMUNICATION AND COMPUTING RESOURCES

The computing resources equipped in communication agents are limited, and can hardly support retrain and fine-tune tasks locally. However, offloading such tasks to cloud servers requires additional communication overhead. Therefore, the joint scheduling of computation and communication resources in SC networks is a bottleneck issue that needs to be overcome. Developing edge intelligence enabled SC architecture is a way to reduce the computing pressure on terminal devices and the communication overhead caused by data sharing. A promising research direction is the task oriented computation offloading strategies. In addition, the aggregation strategy for global SC systems and the node selection strategies for participating in the aggregations is also a key direction to achieve the trade offs between semantic communication performance and resource overhead.

2) JOINTLY SCHEDULE COMMUNICATION AND STORAGE RESOURCES

The caching technologies have been proven to reduce communication latency and resource overhead in traditional communication systems [128]. The existing caching technologies enables the routers, edge nodes, and BSs to cache original data according to user preferences or information popularity, aiming to maximize the probability that the cached data meets the user's request. However, directly applying existing caching technologies to SC systems will lead to two problems. First, repeatedly extracting semantic information from cached data is not beneficial for improving the efficiency of semantic communication. Second, storage resources could be wasted since the data size of the extracted SI is generally much smaller than the original data. A promising research direction is to cache the SI extracted from historical interactions. However, the caching technology for SC systems must consider the SI parsing capabilities of receivers since the SI extracted from the same original data can be represented by multiple data structures. In addition, the prior knowledge tightly coupled with the cache SI should also be cached [129]. Moreover, it is necessary to study cache refresh mechanisms to determine the lifetime of cached SI due to the dynamic context of SCs.

G. JOINT OPTIMIZATION OF SI ENCODING AND SC SECURITY

The security and privacy protection are always the important topics in wireless networks. Since SC technologies extract and encode SI based on KBs or intelligent models shared by communication participants, SC technologies can be considered as a potential method for secure communication [50]. However, compared to standard bit-wise communications

focus on the amount of leaked data, the security issues of SCs focus on the leakage of the meanings of the data and attacks on DL models used for SI extraction. The authors in [130] provided an overview on the fundamentals and key challenges in the design of secure semantic communication. The work in [131] introduced the covert communication technologies to SC networks that makes it difficult for eavesdroppers to determine whether SI has transmitted. To prevent SI leakage, a promising solution is to integrate covert communication technologies [131] with SC systems. To against adversarial attacks on DL models, a promising solution is to involve adversarial examples detection approaches [132] in SC systems.

VI. CONCLUSION

In this paper, we provided a comprehensive survey of SC in future 6G networks. Firstly, we outlined the concepts of SI and SC. Then, starting from major scenarios of future 6G networks, we described the mutually supportive relationship between SC technologies and 6G networks. Next, we reviewed the development of the fundamental theories related to SC as well as four key technologies involved in SC system, such as architecture design, SI extraction, SI transmission, and performance evaluation. We respectively discussed the state-of-art contributions and challenges of each key technologies. Finally, we introduced some future research directions of SC technologies with insights for further in-depth exploration.

REFERENCES

[1] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Netw.*, vol. 34, no. 3, pp. 134–142, May 2020.

[2] J. A. del Peral-Rosado, R. Raulefs, J. A. López-Salcedo, and G. Seco-Granados, "Survey of cellular mobile radio localization methods: From 1G to 5G," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 2, pp. 1124–1148, 2nd Quart., 2018.

[3] *Framework and Overall Objectives of the Future Development of Imt for 2030 and Beyond*, Int. Telecommun. Union (ITU) Recommendation (ITU-R), Geneva, Switzerland, Jun. 2023.

[4] L. Huawei Technologies Co. (2017). *Huawei ILab VR Technology White Paper.Cloud VR Bearer Networks*. [Online]. Available: https://www-file.huawei.com/-/media/corporate/pdf/ilab/cloud vr oriented bearer network white paper en v2.pdf

[5] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, Jul. 1948.

[6] W. Weaver, "Recent contributions to the mathematical theory of communication," *ETC Rev. Gen. Semant.*, vol. 10, no. 4, pp. 261–281, 1953.

[7] X. Luo, H.-H. Chen, and Q. Guo, "Semantic communications: Overview, open issues, and future research directions," *IEEE Wireless Commun.*, vol. 29, no. 1, pp. 210–219, Feb. 2022.

[8] D. Wheeler and B. Natarajan, "Engineering semantic communication: A survey," *IEEE Access*, vol. 11, pp. 13965–13995, 2023.

[9] T. M. Getu, G. Kaddoum, and M. Bennis, "Making sense of meaning: A survey on metrics for semantic and goal-oriented communication," *IEEE Access*, vol. 11, pp. 45456–45492, 2023.

[10] X. Zhang, J. Gu, X. Li, Q. Cui, and X. Tao, "Research and prospect of semantic communications," *Basic Sci. China*, vol. 25, no. 4, pp. 1–7, Aug. 2023.

[11] Z. Qin, T. Zhao, F. Li, and X. Tao, "Survey of research on multimodal semantic communication," *J. Commun.*, vol. 44, no. 5, pp. 28–41, May 2023.

[12] W. Yang, H. Du, Z. Q. Liew, W. Y. B. Lim, Z. Xiong, D. Niyato, X. Chi, X. Shen, and C. Miao, "Semantic communications for future Internet: Fundamentals, applications, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 213–250, 1st Quart., 2023.

[13] E. Calvanese Strinati and S. Barbarossa, "6G networks: Beyond Shannon towards semantic and goal-oriented communications," *Comput. Netw.*, vol. 190, May 2021, Art. no. 107930. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1389128621000773

[14] Y. Liu, X. Wang, Z. Ning, M. Zhou, L. Guo, and B. Jedari, "A survey on semantic communications: Technologies, solutions, applications and challenges," *Digit. Commun. Netw.*, vol. 10, no. 3, pp. 528–545, Jun. 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352864823000925

[15] Y. Zhong, "A theory of semantic information," *China Commun.*, vol. 14, no. 1, pp. 1–17, Jan. 2017.

[16] G. Shi, Y. Xiao, Y. Li, D. Gao, and X. Xie, "Semantic communication networking for the intelligence of everything," *Chin. J. Internet Things*, vol. 5, no. 2, pp. 26–36, Jun. 2021.

[17] D. Gündüz, Z. Qin, I. E. Aguerri, H. S. Dhillon, Z. Yang, A. Yener, K. K. Wong, and C.-B. Chae, "Beyond transmitting bits: Context, semantics, and task-oriented communications," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 5–41, Jan. 2023.

[18] J. Bao, P. Basu, M. Dean, C. Partridge, A. Swami, W. Leland, and J. A. Hendler, "Towards a theory of semantic communication," in *Proc. IEEE Netw. Sci. Workshop*, West Point, NY, USA, Jun. 2011, pp. 110–117.

[19] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, "Deep learning enabled semantic communication systems," *IEEE Trans. Signal Process.*, vol. 69, pp. 2663–2675, 2021.

[20] C. Dong, H. Liang, X. Xu, S. Han, B. Wang, and P. Zhang, "Semantic communication system based on semantic slice models propagation," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 202–213, Jan. 2023.

[21] Y. Wang, M. Chen, T. Luo, W. Saad, D. Niyato, H. V. Poor, and S. Cui, "Performance optimization for semantic communications: An attention-based reinforcement learning approach," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 9, pp. 2598–2613, Sep. 2022.

[22] W. Zhang, Y. Wang, M. Chen, T. Luo, and D. Niyato, "Optimization of image transmission in a cooperative semantic communication networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 2, pp. 861–873, Feb. 2024.

[23] E. Bourtsoulatze, D. Burth Kurka, and D. Gündüz, "Deep joint source-channel coding for wireless image transmission," *IEEE Trans. Cognit. Commun. Netw.*, vol. 5, no. 3, pp. 567–579, Sep. 2019.

[24] C.-H. Lee, J.-W. Lin, P.-H. Chen, and Y.-C. Chang, "Deep learning-constructed joint transmission-recognition for Internet of Things," *IEEE Access*, vol. 7, pp. 76547–76561, 2019.

[25] J. Hu, F. Wang, W. Xu, H. Gao, and P. Zhang, "Scalable multi-task semantic communication system with feature importance ranking," in *Proc. ICASSP - IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Rhodes Island, Greece, Jun. 2023, pp. 1–5.

[26] Q. Pan, H. Tong, J. Lv, T. Luo, Z. Zhang, C. Yin, and J. Li, "Image segmentation semantic communication over Internet of Vehicles," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2023, pp. 1–6.

[27] P. Talli, F. Pase, F. Chiariotti, A. Zanella, and M. Zorzi, "Semantic and effective communication for remote control tasks with dynamic feature compression," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, Hoboken, NJ, USA, May 2023, pp. 1–6.

[28] F. Pase, D. Gündüz, and M. Zorzi, "Rate-constrained remote contextual bandits," *IEEE J. Sel. Areas Inf. Theory*, vol. 3, no. 4, pp. 789–802, Dec. 2022.

[29] J. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson, "Counterfactual multi-agent policy gradients," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, New Orleans, LA, USA, Feb. 2018, pp. 1–12.

[30] B. Juba, *Universal Semantic Communication*. Berlin, Germany: Springer, 2011.

[31] B. Juba and M. Sudan, "Universal semantic communication I," in *Proc.14th Annu. ACM Symp. Theory Comput.*, Victoria, BC, Canada, May 2008, pp. 123–132.

[32] B. Juba and M. Sudan, "Universal semantic communication II: A theory of goal-oriented communication," *Electr. Colloq. Comput. Complex.*, vol. 15, no. 95, pp. 1–29, Feb. 2008.

[33] B. Juba and S. Vempala, "Semantic communication for simple goals is equivalent to on-line learning," in *Proc. 22nd Int. Conf. Algorithmic Learn. Theory*, Espoo, Finland, Jan. 2011, pp. 277–291.

[34] M. K. Farshbafan, W. Saad, and M. Debbah, "Curriculum learning for goal-oriented semantic communications with a common language," *IEEE Trans. Commun.*, vol. 71, no. 3, pp. 1430–1446, Mar. 2023.

[35] L. X. Nguyen, H. Q. Le, Y. L. Tun, P. S. Aung, Y. K. Tun, Z. Han, and C. S. Hong, "An efficient federated learning framework for training semantic communication systems," *IEEE Trans. Veh. Technol.*, vol. 73, no. 10, pp. 15872–15877, Oct. 2024.

[36] H. Sun, H. Tian, W. Ni, and J. Zheng, "Federated learning-based cooperative model training for task-oriented semantic communication," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, Vancouver, BC, Canada, May 2024, pp. 1–6.

[37] J. Xia, T. Liu, Z. Ling, T. Wang, X. Fu, and M. Chen, "PervasiveFL: Pervasive federated learning for heterogeneous IoT systems," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 41, no. 11, pp. 4100–4111, Nov. 2022.

[38] S. Wang, M. Chen, Z. Yang, C. Yin, W. Saad, S. Cui, and H. V. Poor, "Distributed reinforcement learning for age of information minimization in real-time IoT systems," *IEEE J. Sel. Topics Signal Process.*, vol. 16, no. 3, pp. 501–515, Apr. 2022.

[39] Y. Yang, Y. Liu, W. Zhang, and Z. Zhang, "SNR-adaptive deep joint source-channel coding scheme for imagesemantic transmission with convolutional block attention module," *J. China Univ. Posts Telec.*, vol. 31, no. 1, pp. 1–11, 2024.

[40] G. Shi, Y. Xiao, Y. Li, and X. Xie, "From semantic communication to semantic-aware networking: Model, architecture, and open problems," *IEEE Commun. Mag.*, vol. 59, no. 8, pp. 44–50, Aug. 2021.

[41] Y. Bar-Hillel and R. Carnap, "Semantic information," *Brit. J. Philos. Sci.*, vol. 4, no. 14, pp. 147–157, Aug. 1953. [Online]. Available: https://api.semanticscholar.org/CorpusID

[42] J. Hintikka and P. Suppes, *Information and Inference*. Cham, Switzerland: Springer, 1970.

[43] R. Carnap and Y. Bar-Hillel, "An outline of a theory of semantic information," *Br. J. Philos. Sci.*, vol. 19, no. 3, pp. 230–232, Apr. 1954.

[44] S. Sequoiah-Grayson and L. Floridi, "Semantic conceptions of information," in *The Stanford Encyclopedia of Philosophy*, E. N. Zalta, Ed., Metaphysics Research Lab, Stanford University, Spring 2022. [Online]. Available: https://plato.stanford.edu/archives/spr2022/entries/information-semantic/

[45] Y. Yang, C. Guo, F. Liu, C. Liu, L. Sun, Q. Sun, and J. Chen, "Semantic communications with artificial intelligence tasks: Reducing bandwidth requirements and improving artificial intelligence task performance," *IEEE Ind. Electron. Mag.*, vol. 17, no. 3, pp. 2–11, Sep. 2023.

[46] Y. Xiao, Z. Sun, G. Shi, and D. Niyato, "Imitation learning-based implicit semantic-aware communication networks: Multi-layer representation and collaborative reasoning," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 3, pp. 639–658, Mar. 2023.

[47] J. Xu, B. Ai, W. Chen, A. Yang, P. Sun, and M. Rodrigues, "Wireless image transmission using deep source channel coding with attention modules," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 4, pp. 2315–2328, Apr. 2022.

[48] K. Yang, S. Wang, J. Dai, K. Tan, K. Niu, and P. Zhang, "WITT: A wireless image transmission transformer for semantic communications," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Rhodes Island, Greece, Jun. 2023, pp. 1–5.

[49] M. Yang and H.-S. Kim, "Deep joint source-channel coding for wireless image transmission with adaptive rate control," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Singapore, May 2022, pp. 5193–5197.

[50] J. Kang, H. Du, Z. Li, Z. Xiong, S. Ma, D. Niyato, and Y. Li, "Personalized saliency in task-oriented semantic communications: Image transmission and performance analysis," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 186–201, Jan. 2023.

[51] D. B. Kurka and D. Gündüz, "DeepJSCC-f: Deep joint source-channel coding of images with feedback," *IEEE J. Sel. Areas Inf. Theory*, vol. 1, no. 1, pp. 178–193, May 2020.

[52] D. Huang, X. Tao, F. Gao, and J. Lu, "Deep learning-based image semantic coding for semantic communications," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Madrid, Spain, Dec. 2021, pp. 1–6.

[53] Y. Luan, L. He, M. Ostendorf, and H. Hajishirzi, "Multi-task identification of entities, relations, and coreference for scientific knowledge graph construction," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, Brussels, Belgium, Oct. 2018, pp. 1–14. [Online]. Available: https://www.aclweb.org/anthology/D18-1360

[54] H. Xie, Z. Qin, and G. Y. Li, "Task-oriented multi-user semantic communications for VQA," *IEEE Wireless Commun. Lett.*, vol. 11, no. 3, pp. 553–557, Mar. 2022.

[55] D. Huang, F. Gao, X. Tao, Q. Du, and J. Lu, "Toward semantic communications: Deep learning-based image semantic coding," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 55–71, Jan. 2023.

[56] A. Li, X. Liu, G. Wang, and P. Zhang, "Domain knowledge driven semantic communication for image transmission over wireless channels," *IEEE Wireless Commun. Lett.*, vol. 12, no. 1, pp. 55–59, Jan. 2023.

[57] H. Nam, J. Park, J. Choi, and S.-L. Kim, "Sequential semantic generative communication for progressive text-to-image generation," in *Proc. 20th Annu. IEEE Int. Conf. Sens., Commun., Netw. (SECON)*, Sep. 2023, pp. 91–94.

[58] Q. Hu, G. Zhang, Z. Qin, Y. Cai, G. Yu, and G. Y. Li, "Robust semantic communications with masked VQ-VAE enabled codebook," *IEEE Trans. Wireless Commun.*, vol. 22, no. 12, pp. 8707–8722, Dec. 2023.

[59] X. Kang, B. Song, J. Guo, Z. Qin, and F. R. Yu, "Task-oriented image transmission for scene classification in unmanned aerial systems," *IEEE Trans. Commun.*, vol. 70, no. 8, pp. 5181–5192, Aug. 2022.

[60] M. U. Lokumarambage, V. S. S. Gowrisetty, H. Rezaei, T. Sivalingam, N. Rajatheva, and A. Fernando, "Wireless end-to-end image transmission system using semantic communications," *IEEE Access*, vol. 11, pp. 37149–37163, 2023.

[61] W. Tong, F. Liu, Z. Sun, Y. Yang, and C. Guo, "Image semantic communications: An extended rate-distortion theory based scheme," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Rio de Janeiro, Brazil, Dec. 2022, pp. 1723–1728.

[62] Z. Weng and Z. Qin, "Semantic communication systems for speech transmission," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 8, pp. 2434–2444, Aug. 2021.

[63] J. Guo, Y. Zhang, C. Liu, W. Xu, and Z. Bie, "SNR-adaptive multi-layer semantic communication for speech," in *Proc. IEEE 34th Annu. Int. Symp. Pers., Indoor Mobile Radio Commun. (PIMRC)*, Toronto, ON, Canada, Sep. 2023, pp. 1–6.

[64] Z. Xiao, S. Yao, J. Dai, S. Wang, K. Niu, and P. Zhang, "Wireless deep speech semantic transmission," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Rhodes Island, Greece, Jun. 2023, pp. 1–5.

[65] Z. Weng, Z. Qin, and X. Tao, "Semantic-aware speech-to-text transmission over MIMO channels," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Rome, Italy, Jun. 2023, pp. 1362–1367.

[66] Z. Weng, Z. Qin, X. Tao, C. Pan, G. Liu, and G. Y. Li, "Deep learning enabled semantic communications with speech recognition and synthesis," *IEEE Trans. Wireless Commun.*, vol. 22, no. 9, pp. 6227–6240, Sep. 2023.

[67] T. Han, Q. Yang, Z. Shi, S. He, and Z. Zhang, "Semantic-preserved communication system for highly efficient speech transmission," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 245–259, Jan. 2023.

[68] H. Xie and Z. Qin, "A lite distributed semantic communication system for Internet of Things," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 142–153, Jan. 2021.

[69] B. Tang, Q. Li, L. Huang, and Y. Yin, "Text semantic communication systems with sentence-level semantic fidelity," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Glasgow, U.K., Mar. 2023, pp. 1–6.

[70] X. Peng, Z. Qin, D. Huang, X. Tao, J. Lu, G. Liu, and C. Pan, "A robust deep learning enabled semantic communication system for text," in *Proc. IEEE Global Commun. Conf.*, Rio de Janeiro, Brazil, Dec. 2022, pp. 2704–2709.

[71] S. K. Dam, Md. S. Munir, A. D. Raha, A. Adhikary, S.-B. Park, and C. S. Hong, "RNN-based text summarization for communication cost reduction: Toward a semantic communication," in *Proc. Int. Conf. Inf. Netw. (ICOIN)*, Bangkok, Thailand, Jan. 2023, pp. 423–426.

[72] T. Mikolov, K. Chen, G. S. Corrado, and J. B. Dean, "Efficient estimation of word representations in vector space," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Scottsdale, AZ, USA, Jan. 2013, pp. 1–12.

[73] L. Hu, Y. Li, H. Zhang, L. Yuan, F. Zhou, and Q. Wu, "Robust semantic communication driven by knowledge graph," in *Proc. 9th Int. Conf. Internet Things, Syst., Manage. Secur. (IOTSMS)*, Milan, Italy, Nov. 2022, pp. 1–5.

[74] Q. Zhou, R. Li, Z. Zhao, C. Peng, and H. Zhang, "Semantic communication with adaptive universal transformer," *IEEE Wireless Commun. Lett.*, vol. 11, no. 3, pp. 453–457, Mar. 2022.

[75] Y. Sheng, F. Li, L. Liang, and S. Jin, "A multi-task semantic communication system for natural language processing," in *Proc. IEEE 96th Veh. Technol. Conf. (VTC-Fall)*, London, U.K., Sep. 2022, pp. 1–5.

[76] H. Xie, Z. Qin, X. Tao, and K. B. Letaief, "Task-oriented multi-user semantic communications," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 9, pp. 2584–2597, Sep. 2022.

[77] S. R. Pokhrel and J. Choi, "Understand-before-talk (UBT): A semantic communication approach to 6G networks," *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 3544–3556, Mar. 2023.

[78] S. Ji, S. Pan, E. Cambria, P. Marttinen, and P. S. Yu, "A survey on knowledge graphs: Representation, acquisition, and applications," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 494–514, Feb. 2022.

[79] K. Tang, Y. Niu, J. Huang, J. Shi, and H. Zhang, "Unbiased scene graph generation from biased training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 3713–3722.

[80] F. Zhou, Y. Li, M. Xu, L. Yuan, Q. Wu, R. Q. Hu, and N. Al-Dhahir, "Cognitive semantic communication systems driven by knowledge graph: Principle, implementation, and performance evaluation," *IEEE Trans. Commun.*, vol. 72, no. 1, pp. 193–208, Jan. 2024.

[81] S. Kadam and D. In Kim, "Knowledge-aware semantic communication system design," in *Proc. IEEE Int. Conf. Commun.*, Jun. 2023, pp. 6102–6107.

[82] B. Wang, R. Li, J. Zhu, Z. Zhao, and H. Zhang, "Knowledge enhanced semantic communication receiver," *IEEE Commun. Lett.*, vol. 27, no. 7, pp. 1794–1798, Jul. 2023.

[83] Z. Qin Liew, M. Xu, W. Yang Bryan Lim, Z. Xiong, D. Niyato, and D. In Kim, "Mechanism design for semantic communication in UAV-assisted metaverse: A combinatorial auction approach," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2236–2251, Feb. 2024.

[84] Y. Chang, L. Ballotta, and L. Carlone, "D-lite: Navigation-oriented compression of 3D scene graphs for multi-robot collaboration," *IEEE Robot. Autom. Lett.*, vol. 8, no. 11, pp. 7527–7534, Nov. 2023.

[85] C. Li, Y. Duan, Q. Du, C. Pan, G. Liu, and X. Tao, "Image generation from scene graph with object edges," in *Proc. IEEE 96th Veh. Technol. Conf. (VTC-Fall)*, London, U.K., Sep. 2022, pp. 1–7.

[86] Z. Wang, Y. Li, D. Huang, Y. Luo, N. Ge, and J. Lu, "Deformable geometry based semantic reconstruction from scene graphs," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Madrid, Spain, Dec. 2021, pp. 1–6.

[87] Y. Wilks, "An intelligent analyzer and understander of English," *Commun. ACM*, vol. 18, no. 5, pp. 264–274, May 1975, doi: 10.1145/360762.360770.

[88] G. Shi, Y. Li, and X. Xie, "Semantic communications: Outcome of the intelligence era," *Pattern Recognit. Artif. Intell.*, vol. 31, no. 1, pp. 91–99, Jan. 2018.

[89] P. Zhang, W. Xu, H. Gao, K. Niu, X. Xu, X. Qin, C. Yuan, Z. Qin, H. Zhao, J. Wei, and F. Zhang, "Toward wisdom-evolutionary and primitive-concise 6G: A new paradigm of semantic communication networks," *Engineering*, vol. 8, pp. 60–73, Jan. 2022.

[90] Y. Wang, Y. Hu, H. Du, T. Luo, and D. Niyato, "Multi-agent reinforcement learning for covert semantic communications over wireless networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Rhodes Island, Greece, Jun. 2023, pp. 1–5.

[91] G. Zhang, Q. Hu, Z. Qin, Y. Cai, and G. Yu, "A unified multi-task semantic communication system with domain adaptation," in *Proc. IEEE Global Commun. Conf.*, Rio de Janeiro, Brazil, Dec. 2022, pp. 3971–3976.

[92] H. Zhang, S. Shao, M. Tao, X. Bi, and K. B. Letaief, "Deep learning-enabled semantic communication systems with task-unaware transmitter and dynamic data," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 170–185, Jan. 2023.

[93] W. Zhang, H. Zhang, H. Ma, H. Shao, N. Wang, and V. C. M. Leung, "Predictive and adaptive deep coding for wireless image transmission in semantic communication," *IEEE Trans. Wireless Commun.*, vol. 22, no. 8, pp. 5486–5501, Aug. 2023.

[94] J. Shao, Y. Mao, and J. Zhang, "Learning task-oriented communication for edge inference: An information bottleneck approach," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 1, pp. 197–211, Jan. 2022.

[95] J. Dai, S. Wang, K. Tan, Z. Si, X. Qin, K. Niu, and P. Zhang, "Nonlinear transform source-channel coding for semantic communications," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 8, pp. 2300–2316, Aug. 2022.

[96] J. Dai, S. Wang, K. Yang, K. Tan, X. Qin, Z. Si, K. Niu, and P. Zhang, "Toward adaptive semantic communications: Efficient data transmission via online learned nonlinear transform source-channel coding," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 8, pp. 2609–2627, Aug. 2023.

[97] Y. Wang, M. Chen, Z. Yang, W. Saad, T. Luo, S. Cui, and H. V. Poor, "Meta-reinforcement learning for reliable communication in THz/VLC wireless VR networks," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 7778–7793, Sep. 2022.

[98] S. Wang, M. Chen, X. Liu, C. Yin, S. Cui, and H. V. Poor, "A machine learning approach for task and resource allocation in mobile-edge computing-based networks," *IEEE Internet Things J.*, vol. 8, no. 3, pp. 1358–1372, Feb. 2021.

[99] Y. Wang, M. Chen, Z. Yang, T. Luo, and W. Saad, "Deep learning for optimal deployment of UAVs with visible light communications," *IEEE Trans. Wireless Commun.*, vol. 19, no. 11, pp. 7049–7063, Nov. 2020.

[100] H. Zhang, H. Wang, Y. Li, K. Long, and A. Nallanathan, "DRL-driven dynamic resource allocation for task-oriented semantic communication," *IEEE Trans. Commun.*, vol. 71, no. 7, pp. 3992–4004, Jul. 2023.

[101] B. Du, H. Du, H. Liu, D. Niyato, P. Xin, J. Yu, M. Qi, and Y. Tang, "YOLO-based semantic communication with generative AI-aided resource allocation for digital twins construction," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 7664–7678, Mar. 2024.

[102] C. Liu, C. Guo, Y. Yang, and N. Jiang, "Adaptable semantic compression and resource allocation for task-oriented communications," *IEEE Trans. Cognit. Commun. Netw.*, vol. 10, no. 3, pp. 769–782, Jun. 2024.

[103] L. Yan, Z. Qin, R. Zhang, Y. Li, and G. Y. Li, "Resource allocation for text semantic communications," *IEEE Wireless Commun. Lett.*, vol. 11, no. 7, pp. 1394–1398, Jul. 2022.

[104] P. Zhang, X. Xu, C. Dong, K. Niu, H. Liang, Z. Liang, X. Qin, M. Sun, H. Chen, N. Ma, W. Xu, G. Wang, and X. Tao, "Model division multiple access for semantic communications," *Frontiers Inf. Technol. Electron. Eng.*, vol. 24, no. 6, pp. 801–812, May 2023.

[105] H. Liang, K. Liu, X. Liu, H. Jiang, C. Dong, X. Xu, K. Niu, and P. Zhang, "Orthogonal model division multiple access," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 11693–11707, Sep. 2024.

[106] C.-G. Lu, "A generalization of Shannon's information theory," *Int. J. Gen. Syst.*, vol. 28, no. 6, pp. 453–490, Aug. 1999.

[107] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. meeting Assoc. Comput. Linguistics*, Philadelphia, PA, USA, Jul. 2002, pp. 311–318.

[108] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. Acl Workshop Intrinsic Extrinsic Eval. Measures Mach. Transl. Summarization*, Ann Arbor, MI, USA, Jun. 2005, pp. 65–72.

[109] F. Wang, M. Jiang, C. Qian, S. Yang, C. Li, H. Zhang, X. Wang, and X. Tang, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6450–6458.

[110] D. M. W. Powers, "Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation," 2020, *arXiv:2010.16061*.

[111] A. Milan, L. Leal-Taixe, I. Reid, S. Roth, and K. Schindler, "MOT16: A benchmark for multi-object tracking," 2016, *arXiv:1603.00831*.

[112] P. Anderson, B. Fernando, M. Johnson, and S. J. Gould, "SPICE: Semantic propositional image caption evaluation," in *Proc. 14th Eur. Conf. Comput. Vision–ECCV*, Amsterdam, The Netherlands, Jan. 2016, pp. 382–398.

[113] M. Kountouris and N. Pappas, "Semantics-empowered communication for networked intelligent systems," *IEEE Commun. Mag.*, vol. 59, no. 6, pp. 96–102, Jun. 2021.

[114] A. Maatouk, M. Assaad, and A. Ephremides, "The age of incorrect information: An enabler of semantics-empowered communication," *IEEE Trans. Wireless Commun.*, vol. 22, no. 4, pp. 2621–2635, Apr. 2023.

[115] Z. Yang, M. Chen, Z. Zhang, and C. Huang, "Energy efficient semantic communication over wireless networks with rate splitting," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 5, pp. 1484–1495, May 2023.

[116] Z. Zheng, Z. Li, H. Jiang, L. Y. Zhang, and D. Tu, "Semantic-aware privacy-preserving online location trajectory data sharing," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 2256–2271, 2022.

[117] I. V. Bajic, W. Lin, and Y. Tian, "Collaborative intelligence: Challenges and opportunities," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Toronto, ON, Canada, Jun. 2021, pp. 8493–8497.

[118] H. Zimmermann, "OSI reference model–The ISO model of architecture for open systems interconnection," *IEEE Trans. Commun.*, vol. TC-28, no. 4, pp. 425–432, Apr. 1980.

[119] Y. Bo, Y. Duan, S. Shao, and M. Tao, "Learning based joint coding-modulation for digital semantic communication systems," in *Proc. 14th Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Nanjing, China, Nov. 2022, pp. 1–6.

[120] Y. Che, H. Xiong, S. Han, and X. Xu, "Cache-enabled knowledge base construction strategy in semantic communications," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Rio de Janeiro, Brazil, Dec. 2022, pp. 1507–1512.
[121] J. Liang, Y. Xiao, Y. Li, G. Shi, and M. Bennis, "Life-long learning for reasoning-based semantic communication," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, May 2022, pp. 271–276.
[122] M. Chen, D. Gündüz, K. Huang, W. Saad, M. Bennis, A. V. Feljan, and H. V. Poor, "Distributed learning in wireless networks: Recent progress and future challenges," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 12, pp. 3579–3605, Dec. 2021.
[123] M. Chen, Z. Yang, W. Saad, C. Yin, H. V. Poor, and S. Cui, "A joint learning and communications framework for federated learning over wireless networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 269–283, Jan. 2021.
[124] M. Chen, N. Shlezinger, H. V. Poor, Y. C. Eldar, and S. Cui, "Communication-efficient federated learning," *Proc. Natl. Acad. Sci.*, vol. 118, no. 17, Mar. 2021, Art. no. e2024789118.
[125] Y. Xiao, Y. Liao, Y. Li, G. Shi, H. V. Poor, W. Saad, M. Debbah, and M. Bennis, "Reasoning over the air: A reasoning- based implicit semantic-aware communication framework," *IEEE Trans. Wireless Commun.*, vol. 23, no. 4, pp. 3839–3855, Apr. 2024.
[126] H. Xiao, M. Huang, and X. Zhu, "TransG : A generative model for knowledge graph embedding," in *Proc. 54th Annu. Meeting Assoc. Comput. Linguistics*, Berlin, Germany, 2016, pp. 1–12.
[127] L. A. Galárraga, C. Teflioudi, K. Hose, and F. Suchanek, "AMIE: Association rule mining under incomplete evidence in ontological knowledge bases," in *Proc. 22nd Int. Conf. World Wide Web*, Rio de Janeiro, Brazil, May 2013, pp. 413–422.
[128] M. Sheraz, M. Ahmed, X. Hou, Y. Li, D. Jin, Z. Han, and T. Jiang, "Artificial intelligence for wireless caching: Schemes, performance, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 631–661, 1st Quart., 2021.
[129] W. Yu and J. Zhao, "Semantic communications, semantic edge computing, and semantic caching with applications to the metaverse and 6G mobile networks," in *Proc. IEEE 43rd Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Jul. 2023, pp. 983–984.
[130] Z. Yang, M. Chen, G. Li, Y. Yang, and Z. Zhang, "Secure semantic communications: Fundamentals and challenges," *IEEE Netw.*, vol. 38, no. 6, pp. 513–520, Nov. 2024.
[131] M. R. Bloch, "Covert communication over noisy channels: A resolvability perspective," *IEEE Trans. Inf. Theory*, vol. 62, no. 5, pp. 2334–2354, May 2016.
[132] A. Aldahdooh, W. Hamidouche, S. A. Fezza, and O. Déforges, "Adversarial example detection for DNN models: A review and experimental comparison," *Artif. Intell. Rev.*, vol. 55, no. 6, pp. 4403–4462, Jan. 2022.

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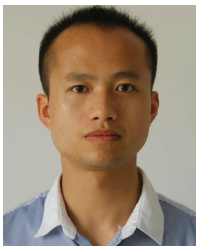

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